Macroeconomics I — MPE Insper

The hardest textbook exercises on Listas 1–7: list and solutions

How to use this. Thirty exercises from Kurlat (2020) and Romer (2012), the hardest the two books offer on the course's topics. Statements are not reproduced: open the book at the page given and attempt the exercise fully before reading the solution. Kurlat's exercises are solved step by step, with verification code, in the existing `Resolucao/kurlat_solutions_*.pdf` (the page column below is the PDF page). Romer's problems are solved in Part II of this document; every number and key derivation there is re-checked by the scripts in `Listas/hardest-parts/`. Weights, readings and the study roadmap are in `Listas/pratica-hardest.md`.

Part I — The list

#	Book	No.	Page	Topic	Level	Solution
1	Kurlat	2.7	42	Welfare: risk aversion	Advanced	`ch1-4`, p. 29
2	Kurlat	2.8	42	Welfare: inequality	Advanced	`ch1-4`, p. 30
3	Kurlat	2.9	43	Welfare: cross-country	Advanced	`ch1-4`, p. 20
4	Kurlat	2.10	43	Welfare: difference principle	Advanced	`ch1-4`, p. 32
5	Kurlat	4.4	73	AK model	Advanced	`ch1-4`, p. 50
6	Kurlat	4.5	73	Malthusian model	Advanced	`ch1-4`, p. 53
7	Kurlat	5.2	94	Speed of convergence	Advanced	`ch05`, p. 7
8	Kurlat	5.7	96	Accounts, returns, growth accounting	Advanced	`ch05`, p. 13
9	Kurlat	5.8	97	Golden Rule from national accounts	Advanced	`ch05`, p. 16
10	Kurlat	6.9	125	Marginal propensity to consume	Advanced	`ch06`, p. 8
11	Kurlat	7.5	148	Prescott's calculation	Advanced	`ch07`, p. 14
12	Kurlat	9.12	185	Optimal vs fixed saving	Advanced	`ch09`, p. 30
13	Kurlat	9.13	186	First Welfare Theorem failure	Advanced	`ch09`, p. 16
14	Kurlat	10.5	203	Baumol–Tobin	Advanced	`ch10-11`, p. 8
15	Kurlat	11.6	219	Seignorage, Laffer curve	Advanced	`ch10-11`, p. 17
16	Kurlat	11.7	221	Bank seignorage	Advanced	`ch10-11`, p. 23
17	Romer	1.12	47–48	Embodied technical progress	Graduate	Part II
18	Romer	1.15	48	Resources and land	Graduate	Part II
19	Romer	4.8	186	Education sector	Graduate	Part II
20	Romer	8.1	398	Life-cycle saving	Graduate	Part II
21	Romer	8.13	402–403	Habit formation	Graduate	Part II
22	Romer	8.15	403–404	Time inconsistency	Graduate	Part II
23	Romer	10.16	512	Search efficiency	Graduate	Part II
24	Romer	2.9	95	Closed-form Ramsey	Graduate	Part II
25	Romer	2.12	96	Temporary capital taxes	Graduate	Part II
26	Romer	11.1	576	Forward-looking Cagan	Graduate	Part II
27	Romer	11.22	583	Adaptive-expectations Cagan	Graduate	Part II
28	Romer	6.16	311	Flexible vs rigid prices	Graduate	Part II
29	Romer	6.10	308–309	Menu costs, multiple equilibria	Graduate	Part II
30	Romer	6.7	308	Liquidity trap	Graduate	Part II

Book pages are printed pages: Kurlat (2020), *A Course in Modern Macroeconomics*; Romer (2012), *Advanced Macroeconomics*, 4th ed. Solution pages for Kurlat refer to the PDF pages of `Resolucao/kurlat_solutions_<chapter>.pdf`.

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Part II — Solutions to the Romer problems

Romer 1.12 — Embodied technological progress

Statement in the book, pp. 47–48. Background: Romer §1.3, §1.5.

Notation and assumptions. $g_X \equiv \dot{X}/X$ for any variable X. Labor grows at rate n and A at rate $\mu > 0$; $0 < \alpha < 1$, $0 < s < 1$ and $n + \delta > 0$. Throughout, $\phi \equiv \alpha/(1-\alpha)$, so that $\alpha = \phi(1-\alpha)$ and $1 + \phi = 1/(1-\alpha)$.

(a)

Step — Capital-augmenting progress is labor-augmenting in disguise. Because $\alpha = \phi(1-\alpha)$, $A^\alpha = (A^\phi)^{1-\alpha}$, and the production function can be rewritten as

$$Y = A^\alpha K^\alpha L^{1-\alpha} = K^\alpha (A^\phi L)^{1-\alpha} \quad \Longrightarrow \quad y \equiv \frac{Y}{A^\phi L} = \Big(\frac{K}{A^\phi L}\Big)^\alpha = k^\alpha, \qquad k \equiv \frac{K}{A^\phi L}.$$

Step — Law of motion of k. Log-differentiating the definition of k gives $g_k = g_K - \phi\mu - n$. Dividing $\dot{K} = sY - \delta K$ by K gives $g_K = s\,Y/K - \delta$, and $Y/K = y/k = k^{\alpha-1}$. Hence

$$\frac{\dot{k}}{k} = s\,k^{\alpha-1} - m_a, \qquad m_a \equiv n + \delta + \phi\mu > 0, \qquad \text{that is,} \qquad \dot{k} = s\,k^\alpha - m_a\,k.$$

Step — Global convergence. The growth rate of k, $\Gamma_a(k) \equiv s\,k^{\alpha-1} - m_a$, satisfies

$$\Gamma_a'(k) = -(1-\alpha)s\,k^{\alpha-2} < 0, \qquad \lim_{k\downarrow 0} \Gamma_a(k) = +\infty, \qquad \lim_{k\to\infty} \Gamma_a(k) = -m_a < 0.$$

So it has a unique zero, $k^* = (s/m_a)^{1/(1-\alpha)}$, with $\dot{k} > 0$ for $k < k^*$ and $\dot{k} < 0$ for $k > k^*$. From any $k(0) > 0$, k moves monotonically toward k^* and cannot cross it (it is a rest point), so $k(t) \to k^*$.

Step — Balanced growth. At $k = k^*$, $y = y^* = (k^*)^\alpha$ is constant, so $K = k^* A^\phi L$ and $Y = y^* A^\phi L$ both grow at the growth rate of $A^\phi L$:

$$g_K^{\text{bgp}} = g_Y^{\text{bgp}} = n + \phi\mu = n + \frac{\alpha\mu}{1-\alpha}, \qquad g_{Y/L}^{\text{bgp}} = g_{K/L}^{\text{bgp}} = \frac{\alpha\mu}{1-\alpha}.$$

Economic intuition. With Cobb–Douglas technology it does not matter which factor A multiplies: A^α on capital is $(A^\phi)^{1-\alpha}$ on labor, so capital-augmenting progress at rate μ is labor-augmenting progress at rate $\phi\mu$, and the Solow result (output per worker grows at the rate of labor-augmenting progress) delivers $\alpha\mu/(1-\alpha)$. That is why the problem imposes Cobb–Douglas: otherwise capital-augmenting progress is incompatible with balanced growth (Uzawa's theorem).

(b)

Step — Change of variable. Let $\tilde{J} \equiv J/A$. Then $Y = J^{\alpha}L^{1-\alpha} = (A\tilde{J})^{\alpha}L^{1-\alpha}$, the technology of (a) with $\tilde{J}$ in place of K. As in (a), $Y = \tilde{J}^{\alpha}(A^{\phi}L)^{1-\alpha}$, so $y \equiv Y/(A^{\phi}L) = j^{\alpha}$ with $j \equiv \tilde{J}/(A^{\phi}L)$.

Step — Accumulation of $\tilde{J}$: obsolescence. Using $\dot{J} = sAY - \delta J$ and $J = A\tilde{J}$,

$$g_{\tilde{J}} = g_J - \mu = \frac{sAY}{J} - \delta - \mu = \frac{sY}{\tilde{J}} - (\delta + \mu) \quad \Longrightarrow \quad \dot{\tilde{J}} = sY - (\delta + \mu)\tilde{J}.$$

This is the capital accumulation equation of (a) with depreciation rate $\delta + \mu$. Integrating,

$$\tilde{J}(t) = e^{-(\delta+\mu)t}\,\tilde{J}(0) + \int_0^t e^{-(\delta+\mu)(t-v)}\,sY(v)\,dv :$$

one unit of output invested at v still supplies $e^{-\delta(t-v)}A(v)$ units of J at t, as many as $e^{-(\delta+\mu)(t-v)}$ units of output invested at t would. So $\tilde{J}$ measures the stock in current-vintage machines, and it loses value through decay (δ) and obsolescence (μ).

Step — Law of motion of j. Log-differentiating, $g_j = g_{\tilde{J}} - \phi\mu - n$. With $Y/\tilde{J} = y/j = j^{\alpha-1}$ and $\mu + \phi\mu = \mu/(1-\alpha)$,

$$\dot{j} = s\,j^{\alpha} - m\,j, \qquad m \equiv n + \delta + \frac{\mu}{1-\alpha} = m_a + \mu.$$

Step — Global convergence. The argument of (a) applies with m in place of m_a: $\Gamma(j) \equiv s\,j^{\alpha-1} - m$ has $\Gamma'(j) = -(1-\alpha)s\,j^{\alpha-2} < 0$ and falls from $+\infty$ to $-m < 0$, so from any $j(0) > 0$, j converges monotonically to $j^* = (s/m)^{1/(1-\alpha)}$. The dynamics even solve in closed form: $z \equiv j^{1-\alpha} = j/y = \tilde{J}/Y$ satisfies $\dot{z} = (1-\alpha)j^{-\alpha}\dot{j} = (1-\alpha)(s - mz)$, which is linear, so

$$\frac{\tilde{J}(t)}{Y(t)} = \frac{s}{m} + \Big[\frac{\tilde{J}(0)}{Y(0)} - \frac{s}{m}\Big]e^{-(1-\alpha)m\,t}.$$

Step — Balanced growth. With $j = j^*$ and $y = y^*$ constant, $Y = y^*A^{\phi}L$ and $\tilde{J} = j^*A^{\phi}L$ grow at $n + \phi\mu$, and $J = A\tilde{J}$ grows μ faster:

$$g_Y^{\text{bgp}} = n + \frac{\alpha\mu}{1-\alpha}, \quad g_J^{\text{bgp}} = n + \frac{\mu}{1-\alpha}, \quad g_{Y/L}^{\text{bgp}} = \frac{\alpha\mu}{1-\alpha}, \quad g_{J/Y}^{\text{bgp}} = \mu.$$

Economic intuition. Relative to (a), embodiment changes one thing: the stock that matters, $\tilde{J}$, depreciates at $\delta + \mu$ instead of δ. Depreciation moves levels, not balanced-growth rates, so output grows exactly as in (a). Effective capital J outgrows output by μ because every vintage is more productive than the one before.

(c)

Step — Output on the balanced growth path. On the path, $Y(t) = A(t)^{\phi}L(t)\,y^*$ with $y^* = (j^*)^{\alpha} = (s/m)^{\alpha/(1-\alpha)}$. The paths of A and L and the break-even rate m do not depend on s, so

$$\ln Y(t) = \phi \ln A(t) + \ln L(t) + \frac{\alpha}{1-\alpha}\,(\ln s - \ln m) \quad \Longrightarrow \quad \frac{\partial \ln Y(t)}{\partial \ln s} = \frac{\alpha}{1-\alpha},$$

which is $1/2$ for $\alpha = 1/3$, whatever the values of μ, n and δ.

(d)

Step — Linearize around j^*. Write $\dot{j} = h(j) \equiv s\,j^{\alpha} - m\,j$. At the rest point $h(j^*) = 0$, i.e. $s(j^*)^{\alpha-1} = m$, so $h'(j^*) = \alpha s(j^*)^{\alpha-1} - m = \alpha m - m$. A first-order Taylor expansion gives

$$\dot{j} \simeq -\lambda\,(j - j^*), \qquad \lambda \equiv -h'(j^*) = (1-\alpha)\,m = (1-\alpha)(n+\delta) + \mu,$$

so $j(t) - j^* \simeq e^{-\lambda t}[j(0) - j^*]$.

Step — Output converges at the same rate. Since $y = j^{\alpha}$, $\dot{y} = \alpha j^{\alpha-1}h(j)$, whose derivative with respect to j at j^* is $\alpha(j^*)^{\alpha-1}h'(j^*)$ because $h(j^*) = 0$; and $y - y^* \simeq \alpha(j^*)^{\alpha-1}(j - j^*)$. Hence

$\dot{y} \simeq -\lambda(y - y^*)$ and $y(t) - y^* \simeq e^{-\lambda t}[y(0) - y^*]$. The closed form in (b) shows that $\tilde{J}/Y$ converges at exactly this rate globally, not only near the path.

Step — Magnitude. With $\alpha = 1/3$, $n = 0.01$, $\delta = 0.03$ and $\mu = 0.04$ (so that $g^{\text{bgp}}_{Y/L} = \phi\mu = 0.02$), $\lambda = \frac{2}{3}(0.04) + 0.04 \approx 0.067$ per year: the economy closes about 6.7% of the gap each year, a half-life of $\ln 2/\lambda \approx 10.4$ years.

(e)

Step — Elasticity: unchanged. In the basic model, (1.27) gives $\alpha_K(k^*)/[1 - \alpha_K(k^*)]$, which under Cobb–Douglas is $\alpha/(1-\alpha)$, the same as in (c). In both models balanced-growth output is proportional to $(s/\text{break-even rate})^{\alpha/(1-\alpha)}$, and neither break-even rate ($n + g + \delta$ there, m here) depends on s.

Step — Speed: faster. In the basic model, (1.31) gives $\lambda_S = (1-\alpha)(n + g + \delta)$. Comparing requires a normalization. Holding fixed the balanced-growth rate of output per worker ($g = \phi\mu$), $\lambda_S = (1-\alpha)(n+\delta) + \alpha\mu = (1-\alpha)m_a$, which is also the speed in (a) (linearize as in (d) with m_a), and

$$\lambda - \lambda_S = (1-\alpha)(m - m_a) = (1-\alpha)\mu > 0.$$

Holding fixed the rate of technical progress instead ($g = \mu$), $\lambda - \lambda_S = \alpha\mu > 0$. Either way embodiment speeds up convergence. With the numbers of (d) and $g = \phi\mu = 0.02$, $\lambda_S = \frac{2}{3}(0.06) = 0.04$, the text's 4% with a half-life of about 17 years, against $\lambda \approx 0.067$ and about 10 years.

Economic intuition. The speed of convergence is $1 - \alpha$ times the break-even rate, the rate at which the accumulated stock must be replenished just to keep pace. Embodiment raises that rate by the obsolescence rate μ: old capital is "forgotten" faster, so initial conditions die out faster. The long-run response to s, in contrast, depends only on the curvature α, because s scales gross investment and leaves the break-even rate unchanged. Embodiment matters for transitions, not for the long-run level effect of saving.

Where this goes wrong. Dividing J by A adds $-\mu$ to the law of motion of $\tilde{J}$. Dropping it collapses (b) into (a), gives $\lambda = (1-\alpha)(n+\delta) + \alpha\mu$ and hides the speed-up. Also, the elasticity in (c) is that of output: j^*, and hence J, has elasticity $1/(1-\alpha)$ with respect to s.

Ref: Romer (2012), §1.3 (eq. 1.18) and §1.5 (eqs. 1.27 and 1.31); Solow (1960) and Sato (1966), as cited in the problem.

Checked: `check-1-12-1-15.py` re-derives with sympy $y = k^\alpha$ and $y = j^\alpha$, the laws of motion of k and j, $m = m_a + \mu$, j^*, the elasticity, $h'(j^*)$, the speed of y, the closed form for $\tilde{J}/Y$ and both comparisons with λ_S; it then integrates the levels of K and J numerically and recovers the balanced-growth rates, the elasticity and the speeds $(1-\alpha)m_a$ and $(1-\alpha)m$. The solution manual agrees on every result; its comparison in (e) implicitly sets $g = \mu$. **Romer 1.15**

— Resources and land: deriving equation (1.50)

Statement in the book, p. 48. Background: Romer §1.8, pp. 39–41.

Setting. Production is (1.41), $Y = K^\alpha R^\beta T^\gamma (AL)^{1-\alpha-\beta-\gamma}$ with $\alpha, \beta, \gamma > 0$ and $\alpha + \beta + \gamma < 1$; $\dot{K} = sY - \delta K$, $\dot{L} = nL$ and $\dot{A} = gA$, with $n + \delta > 0$ and $g \geq 0$. Equation (1.50) describes the hypothetical economy of §1.8 in which (1.42)–(1.43) are replaced by $\dot{T} = nT$ and $\dot{R} = nR$: land and resources grow with population. A tilde marks this economy's balanced-growth values, and $g_X \equiv \dot{X}/X$.

Step — A balanced growth path requires $g_Y = g_K$. Dividing $\dot{K} = sY - \delta K$ by K gives (1.44), $g_K = s\,(Y/K) - \delta$. With s and δ constant, g_K is constant if and only if Y/K is constant, that is, if and only if $g_Y = g_K$.

Step — Log-differentiate the production function. Taking logs of (1.41),

$$\ln Y(t) = \alpha \ln K(t) + \beta \ln R(t) + \gamma \ln T(t) + (1-\alpha-\beta-\gamma)[\ln A(t) + \ln L(t)],$$

and differentiating with respect to time (the time derivative of a log is a growth rate) gives (1.46):

$$g_Y(t) = \alpha g_K(t) + \beta g_R(t) + \gamma g_T(t) + (1-\alpha-\beta-\gamma)[g_A(t) + g_L(t)].$$

Step — Insert this economy's growth rates (analog of 1.47). Here $g_R = g_T = g_L = n$ and $g_A = g$, so

$$g_Y(t) = \alpha g_K(t) + (\beta+\gamma)n + (1-\alpha-\beta-\gamma)(n+g).$$

The n terms add up to $(\beta+\gamma)n + (1-\alpha-\beta-\gamma)n = (1-\alpha)n$, hence

$$g_Y(t) = \alpha g_K(t) + (1-\alpha)n + (1-\alpha-\beta-\gamma)g. \tag{1.47$'$}$$

Step — Impose $g_K = g_Y$ (analog of 1.48). On the balanced growth path $g_K = g_Y = \tilde{g}_Y^{\text{bgp}}$. Substituting into (1.47′) and moving $\alpha\tilde{g}_Y^{\text{bgp}}$ to the left gives $(1-\alpha)\tilde{g}_Y^{\text{bgp}} = (1-\alpha)n + (1-\alpha-\beta-\gamma)g$, so

$$\tilde{g}_Y^{\text{bgp}} = \frac{(1-\alpha)n + (1-\alpha-\beta-\gamma)g}{1-\alpha} = n + \frac{1-\alpha-\beta-\gamma}{1-\alpha}\,g. \tag{1.48$'$}$$

Step — Output per worker: equation (1.50). Since $g_L = n$, $\tilde{g}_{Y/L}^{\text{bgp}} = \tilde{g}_Y^{\text{bgp}} - n$, and (1.48′) gives

$$\tilde{g}_{Y/L}^{\text{bgp}} = \frac{1}{1-\alpha}\,(1-\alpha-\beta-\gamma)\,g. \tag{1.50}$$

Step — Convergence (the text's argument, made explicit). Differentiating (1.44) and using $\frac{d}{dt}(Y/K) = (Y/K)(g_Y - g_K)$ gives $\dot{g}_K = s\,(Y/K)(g_Y - g_K) = (g_K+\delta)(g_Y - g_K)$. By (1.47′) and (1.48′), $g_Y - g_K = (1-\alpha)(\tilde{g}_Y^{\text{bgp}} - g_K)$, so

$$\dot{g}_K = (1-\alpha)\,(g_K+\delta)\,(\tilde{g}_Y^{\text{bgp}} - g_K).$$

Since $g_K + \delta = sY/K > 0$, $\dot{g}_K$ has the sign of $\tilde{g}_Y^{\text{bgp}} - g_K$: g_K falls when it is above its balanced-growth value, rises when below, and converges to it monotonically. The subtlety of footnote 23 cannot arise, because the balanced-growth value of Y/K is positive:

$$\frac{Y}{K}\Big|_{\text{bgp}} = \frac{\tilde{g}_Y^{\text{bgp}} + \delta}{s}, \qquad (1-\alpha)(\tilde{g}_Y^{\text{bgp}} + \delta) = (1-\alpha)(n+\delta) + (1-\alpha-\beta-\gamma)g > 0.$$

Linearizing at the rest point, $\partial\dot{g}_K/\partial g_K = -(1-\alpha)(\tilde{g}_Y^{\text{bgp}} + \delta)$, so the speed of convergence is

$$\tilde{\lambda} = (1-\alpha)(n+\delta) + (1-\alpha-\beta-\gamma)\,g.$$

The rate is exact, not only local, for the capital-output ratio, whose law of motion is linear:

$$\frac{d}{dt}\frac{K}{Y} = \frac{K}{Y}\,(g_K - g_Y) = (1-\alpha)\Big[s - (\tilde{g}_Y^{\text{bgp}} + \delta)\frac{K}{Y}\Big].$$

Step — Cross-check: a Solow economy in disguise. With R, T and L all growing at n, the ratios R/L and T/L are constant. Write $R^\beta T^\gamma = (R/L)^\beta (T/L)^\gamma L^{\beta+\gamma}$ and add the exponents of L, $(\beta+\gamma) + (1-\alpha-\beta-\gamma) = 1-\alpha$:

$$R^\beta T^\gamma (AL)^{1-\alpha-\beta-\gamma} = \Big(\frac{R}{L}\Big)^\beta \Big(\frac{T}{L}\Big)^\gamma A^{1-\alpha-\beta-\gamma} L^{1-\alpha} \equiv (\tilde{A}L)^{1-\alpha},$$
$$\tilde{A} \equiv \Big[\Big(\frac{R}{L}\Big)^\beta \Big(\frac{T}{L}\Big)^\gamma\Big]^{1/(1-\alpha)} A^{(1-\alpha-\beta-\gamma)/(1-\alpha)}.$$

So $Y = K^\alpha(\tilde{A}L)^{1-\alpha}$: the basic Cobb–Douglas Solow model with labor-augmenting progress at rate $\tilde{g} \equiv g_{\tilde{A}} = (1-\alpha-\beta-\gamma)g/(1-\alpha)$. On its balanced growth path output per worker grows at $\tilde{g}$, which is (1.50) again, and (1.31) gives the speed $(1-\alpha)(n+\tilde{g}+\delta) = \tilde{\lambda}$.

Step — The growth drag (1.51) follows. Subtracting (1.49) from (1.50),

$$\frac{(1-\alpha-\beta-\gamma)g - [(1-\alpha-\beta-\gamma)g - \beta b - (\beta+\gamma)n]}{1-\alpha} = \frac{\beta b + (\beta+\gamma)n}{1-\alpha},$$

which is (1.51).

Economic intuition. Because the exponents in (1.41) sum to one, (1.46) can be written per worker:

$$g_{Y/L} = \alpha g_{K/L} + \beta g_{R/L} + \gamma g_{T/L} + (1-\alpha-\beta-\gamma)g.$$

Technology multiplies labor only, so with every input per worker held fixed it raises output per worker at the TFP rate $(1-\alpha-\beta-\gamma)g$. Capital is the only reproducible input: on a balanced growth path $g_{K/L} = g_{Y/L}$, which multiplies the TFP rate by $1/(1-\alpha)$. Here $g_{R/L} = g_{T/L} = 0$, so land and resources neither drag nor amplify, and (1.50) isolates technology plus capital deepening. In the actual economy $g_{R/L} = -(b+n)$ and $g_{T/L} = -n$, so the drag (1.51) is the share-weighted decline $\beta(b+n) + \gamma n$ times the same multiplier $1/(1-\alpha)$.

Where this goes wrong. Three slips recur. Using $g_R = -b$ and $g_T = 0$ (the actual economy) reproduces (1.48)–(1.49), not (1.50). Dropping the growth of R and T from g_Y gives

$$\frac{(1-\alpha-\beta-\gamma)g - (\beta+\gamma)n}{1-\alpha},$$

which is (1.49) with $b = 0$, the case of fixed R and T. And the answer is not g, as in the basic model: A augments only labor, whose exponent here is $1-\alpha-\beta-\gamma$, not $1-\alpha$.

Ref: Romer (2012), §1.8, pp. 39–41: eqs. (1.41)–(1.51) and footnote 23; §1.5, eq. (1.31); Nordhaus (1992).

Checked: `check-1-12-1-15.py` re-derives (1.47′), (1.48′) and (1.50) with sympy, recovers the text's (1.48), (1.49) and (1.51) from the same growth-rate equation with $g_R = -b$ and $g_T = 0$, and verifies the law of motion of g_K, its eigenvalue $-\tilde{\lambda}$, the linear equation for K/Y and the $\tilde{A}$ rewriting; it then integrates both economies numerically and recovers (1.50), (1.49), the drag and the speed $\tilde{\lambda}$. The solution manual derives (1.50) by the same steps and agrees; it does not discuss convergence.

Romer 4.8 — Physical and human capital with an education sector

Statement in the book, p. 186. Background: Romer §4.1.

Notation. $x_K \equiv n+g+\delta_K$, $x_H \equiv n+g+\delta_H$, $c_K \equiv s(1-a_K)^{\alpha}(1-a_H)^{1-\alpha}$ and $c_H \equiv B\,a_K^{\gamma}a_H^{\phi}$. L counts people (bodies), so "per person" means per unit of L. Throughout, $0<\alpha<1$, $0<a_K, a_H<1$, $\gamma,\phi>0$ and $\gamma+\phi<1$.

(a)

Step — growth-rate identity. Take logs of $k = K/(AL)$, $\ln k = \ln K - \ln A - \ln L$, and differentiate with respect to time, using $\dot{A}/A = g$ and $\dot{L}/L = n$:

$$\frac{\dot{k}}{k} = \frac{\dot{K}}{K} - g - n \quad\Longrightarrow\quad \dot{k} = \frac{\dot{K}}{AL} - (n+g)\,k; \qquad \text{likewise} \qquad \dot{h} = \frac{\dot{H}}{AL} - (n+g)\,h.$$

Step — goods per unit of effective labor. The exponents of the goods technology sum to one, so dividing by $AL = (AL)^{\alpha}(AL)^{1-\alpha}$,

$$y \equiv \frac{Y}{AL} = \Big[(1-a_K)\frac{K}{AL}\Big]^{\alpha}\Big[(1-a_H)\frac{H}{AL}\Big]^{1-\alpha} = (1-a_K)^{\alpha}(1-a_H)^{1-\alpha}\,k^{\alpha}h^{1-\alpha}.$$

Step — law of motion of k. $\dot{K}/(AL) = sY/(AL) - \delta_K K/(AL) = sy - \delta_K k$; substituting into the identity,

$$\dot{k} = c_K\,k^{\alpha}h^{1-\alpha} - x_K\,k.$$

Step — law of motion of h. The education technology is homogeneous of degree one in (K, H, AL); dividing it by $AL = (AL)^{\gamma}(AL)^{\phi}(AL)^{1-\gamma-\phi}$,

$$\frac{\dot{H}}{AL} + \delta_H h = B\,a_K^{\gamma}a_H^{\phi}\Big(\frac{K}{AL}\Big)^{\gamma}\Big(\frac{H}{AL}\Big)^{\phi}\Big(\frac{AL}{AL}\Big)^{1-\gamma-\phi} = c_H\,k^{\gamma}h^{\phi},$$
$$\dot{h} = c_H\,k^{\gamma}h^{\phi} - x_H\,h.$$

(b)

Step — the $\dot{k}=0$ locus. For $k>0$, $\dot{k}=0$ iff $c_K k^{\alpha-1}h^{1-\alpha}=x_K$, i.e. $(h/k)^{1-\alpha}=x_K/c_K$. Raising both sides to the power $-1/(1-\alpha)$:

$$\dot{k}=0 \iff k=\kappa\, h, \qquad \kappa \equiv \Big(\frac{c_K}{x_K}\Big)^{1/(1-\alpha)}.$$

In (h,k) space (h on the horizontal axis) this is a ray through the origin with slope $\kappa>0$: goods production has constant returns in K and H, so only the ratio h/k matters for $\dot{k}/k$. Off the locus, $\dot{k}=k\,[c_K(h/k)^{1-\alpha}-x_K]$, and for given h the bracket falls with k: $\dot{k}<0$ above the ray, $\dot{k}>0$ below it.

Step — the $\dot{h}=0$ locus. For $h>0$, $\dot{h}=0$ iff $c_H k^{\gamma}=x_H h^{1-\phi}$; raising to the power $1/\gamma$,

$$\dot{h}=0 \iff k=\lambda\, h^{\eta}, \qquad \lambda \equiv \Big(\frac{x_H}{c_H}\Big)^{1/\gamma}, \qquad \eta \equiv \frac{1-\phi}{\gamma}>1,$$

where $\eta>1$ is exactly the assumption $\gamma+\phi<1$. Differentiating,

$$\frac{dk}{dh}=\eta\lambda h^{\eta-1}>0, \qquad \frac{d^2k}{dh^2}=\eta(\eta-1)\lambda h^{\eta-2}>0,$$

so the locus leaves the origin with zero slope ($\eta-1>0$), is increasing and strictly convex, and its slope tends to infinity as $h\to\infty$; equivalently, $h=(c_H k^{\gamma}/x_H)^{1/(1-\phi)}$ is a concave function of k, since $\gamma/(1-\phi)<1$. Off the locus, $\dot{h}=h\,[c_H k^{\gamma}h^{\phi-1}-x_H]$ rises with k: $\dot{h}>0$ above the curve, $\dot{h}<0$ below it.

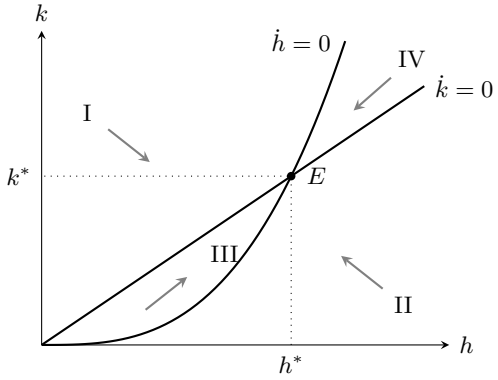


Figure 4.8-1. The $\dot{k}=0$ ray, the convex $\dot{h}=0$ locus, and the direction of motion in each region.

(c)

Step — existence and uniqueness. A balanced growth path is a rest point with $k,h>0$, i.e. a point on both loci: $\kappa h=\lambda h^{\eta} \iff h^{\eta-1}=\kappa/\lambda$. As $\eta-1=(1-\gamma-\phi)/\gamma>0$, $h^{\eta-1}$ rises strictly from 0 to ∞, so there is exactly one positive solution,

$$h^*=\Big(\frac{\kappa}{\lambda}\Big)^{\frac{\gamma}{1-\gamma-\phi}}=\Big(\frac{c_K}{x_K}\Big)^{\frac{\gamma}{(1-\alpha)(1-\gamma-\phi)}}\Big(\frac{c_H}{x_H}\Big)^{\frac{1}{1-\gamma-\phi}},$$
$$k^*=\kappa h^*=\Big(\frac{c_K}{x_K}\Big)^{\frac{1-\phi}{(1-\alpha)(1-\gamma-\phi)}}\Big(\frac{c_H}{x_H}\Big)^{\frac{1}{1-\gamma-\phi}},$$

where the exponent of c_K/x_K in k^* is $\frac{1}{1-\alpha}+\frac{\gamma}{(1-\alpha)(1-\gamma-\phi)}=\frac{1-\phi}{(1-\alpha)(1-\gamma-\phi)}$. Since $\eta>1$, the ray lies above the curve for $0<h<h^*$ and below it for $h>h^*$. The origin is the only other rest point, and the next step shows it attracts no path with $k_0,h_0>0$.

Step — global stability: phase diagram. The loci cut the positive quadrant into four regions (Figure 4.8-1): I above both ($\dot{h}>0$, $\dot{k}<0$), II below both ($\dot{h}<0$, $\dot{k}>0$), III between them with $h<h^*$ ($\dot{h}>0$, $\dot{k}>0$), IV between them with $h>h^*$ ($\dot{h}<0$, $\dot{k}<0$). On the ray bounding III,

$\dot{k}=0$ and $\dot{h}>0$, so the flow points right, into III; on the curve bounding III, $\dot{h}=0$ and $\dot{k}>0$, so it points up, into III. Hence a path never leaves III, and likewise IV. In III, k and h rise and stay below k^*, h^*; in IV they fall and stay above them; a monotone bounded path converges, and its limit must be a rest point: E. In I, h rises and k falls with $\lambda h_0^\eta \le \lambda h^\eta < k \le k_0$ and $h < k/\kappa \le k_0/\kappa$: the path either enters III or IV or stays in I bounded away from 0 and ∞, and converges to E in both cases. II is symmetric ($k_0 \le k < \kappa h \le \kappa h_0$ and $h > (k_0/\lambda)^{1/\eta}$). So every path with $k_0, h_0 > 0$ converges to E: the balanced growth path exists, is unique, and is globally stable.

Step — local stability: Jacobian. Using the locus conditions $c_K k^{\alpha-1}h^{1-\alpha} = x_K$ and $c_H k^\gamma h^{\phi-1} = x_H$ at E,

$$\frac{\partial \dot{k}}{\partial k} = \alpha c_K k^{\alpha-1}h^{1-\alpha} - x_K = -(1-\alpha)x_K, \quad \frac{\partial \dot{k}}{\partial h} = (1-\alpha)c_K k^{\alpha-1}h^{1-\alpha}\,\frac{k}{h} = (1-\alpha)x_K\frac{k^*}{h^*},$$
$$\frac{\partial \dot{h}}{\partial k} = \gamma c_H k^\gamma h^{\phi-1}\,\frac{h}{k} = \gamma x_H \frac{h^*}{k^*}, \qquad \frac{\partial \dot{h}}{\partial h} = \phi c_H k^\gamma h^{\phi-1} - x_H = -(1-\phi)x_H.$$

Hence, since the ratios k^*/h^* cancel in the determinant,

$$\begin{aligned}
\operatorname{tr} J &= -(1-\alpha)x_K - (1-\phi)x_H < 0,\\
\det J &= (1-\alpha)(1-\phi)x_K x_H - (1-\alpha)\gamma\, x_K x_H = (1-\alpha)(1-\gamma-\phi)\, x_K x_H > 0,\\
(\operatorname{tr} J)^2 - 4\det J &= \big[(1-\alpha)x_K + (1-\phi)x_H\big]^2 - 4(1-\alpha)(1-\phi)x_K x_H + 4(1-\alpha)\gamma\, x_K x_H\\
&= \big[(1-\alpha)x_K - (1-\phi)x_H\big]^2 + 4(1-\alpha)\gamma\, x_K x_H > 0.
\end{aligned}$$

Two real negative eigenvalues: E is a stable node (no oscillations), and the eigenvalue closer to zero sets the asymptotic speed of convergence. Again $\det J > 0$ is the assumption $\gamma + \phi < 1$.

Step — growth rates on the balanced path. With $k = k^*$ and $h = h^*$ constant, $K = k^*AL$, $H = h^*AL$ and $Y = y^*AL$ all grow at $n+g$, and per person

$$\frac{K}{L} = A\,k^*, \qquad \frac{H}{L} = A\,h^*, \qquad \frac{Y}{L} = A\,y^*, \qquad y^* = (1-a_K)^\alpha (1-a_H)^{1-\alpha} k^{*\alpha} h^{*1-\alpha},$$

so output, physical capital and human capital per person all grow at rate g.

Economic intuition. K helps produce H and H helps produce goods, hence K; but the feedback dies out because education also uses AL, which cannot be accumulated, and $\gamma + \phi < 1$ means diminishing returns to K and H together. As in Solow, long-run growth per person is the exogenous g.

Where this goes wrong. The condition $\gamma + \phi < 1$ does all the work. With $\gamma + \phi = 1$, $\eta = 1$ and the $\dot{h} = 0$ locus is also a ray; the two rays generically meet only at the origin, there is no rest point with $k, h > 0$, and K and H grow at a rate that depends on s, a_K, a_H and B: an endogenous-growth model (compare Problem 4.9).

(d)

Step — which locus moves. s enters only c_K, hence only $\kappa = (c_K/x_K)^{1/(1-\alpha)}$: a permanent rise from s to $s' > s$ makes the $\dot{k} = 0$ ray steeper and leaves the $\dot{h} = 0$ curve unchanged. The new intersection E' has higher h^* and k^*, since both exponents of c_K in part (c) are positive.

Step — the transition. k and h are stocks, so at the date of the change, t_0, the economy is still at E, which is on the unchanged $\dot{h} = 0$ curve and now strictly below the new ray:

$$\dot{h}(t_0) = 0, \qquad \frac{\dot{k}(t_0)}{k^*} = \frac{s'}{s}\,c_K k^{*\alpha-1}h^{*1-\alpha} - x_K = \Big(\frac{s'}{s} - 1\Big)x_K > 0.$$

The economy enters region III of the new diagram and never leaves it: k and h rise monotonically and converge to E', asymptotically rather than at a finite date. At first only k moves (the path starts vertically); then h rises as more K flows into schooling (Figure 4.8-2, left).

Step — output per person. $\ln(Y/L) = \ln A + \ln y$, with $\ln y$ equal to a constant plus $\alpha \ln k + (1-\alpha)\ln h$, so

$$\frac{d\ln(Y/L)}{dt} = g + \alpha\frac{\dot{k}}{k} + (1-\alpha)\frac{\dot{h}}{h}.$$

At t_0 the level does not jump (k and h are predetermined); its growth rate jumps from g to $g+\alpha(s'/s-1)x_K$, stays above g throughout the transition ($\dot{k},\dot{h}\geq 0$, not both zero), and returns to g as E' is approached. The decline need not be monotone. At t_0, where $\dot{h}=0$,

$$\frac{d}{dt}\frac{\dot{k}}{k} = (1-\alpha)\frac{s'}{s}c_K\Big(\frac{h}{k}\Big)^{1-\alpha}\Big(\frac{\dot{h}}{h}-\frac{\dot{k}}{k}\Big) = -(1-\alpha)\frac{s'}{s}x_K\,\frac{\dot{k}}{k},$$
$$\frac{d}{dt}\frac{\dot{h}}{h} = c_H k^{\gamma}h^{\phi-1}\Big(\gamma\frac{\dot{k}}{k}-(1-\phi)\frac{\dot{h}}{h}\Big) = \gamma x_H\,\frac{\dot{k}}{k},$$
$$\frac{d}{dt}\Big[\alpha\frac{\dot{k}}{k}+(1-\alpha)\frac{\dot{h}}{h}\Big]_{t_0} = (1-\alpha)\Big(\frac{s'}{s}-1\Big)x_K\Big[\gamma x_H-\alpha\frac{s'}{s}x_K\Big],$$

so growth first accelerates when $\gamma x_H > \alpha(s'/s)x_K$: the education channel kicks in with a lag. In the long run $\ln(Y/L)$ is higher, by the elasticity below times the change in $\ln s$, and grows at g again:

$$\frac{d\ln y^*}{d\ln s} = \alpha\frac{1-\phi}{(1-\alpha)(1-\gamma-\phi)}+(1-\alpha)\frac{\gamma}{(1-\alpha)(1-\gamma-\phi)} = \frac{\alpha}{1-\alpha}+\frac{\gamma}{(1-\alpha)(1-\gamma-\phi)}.$$

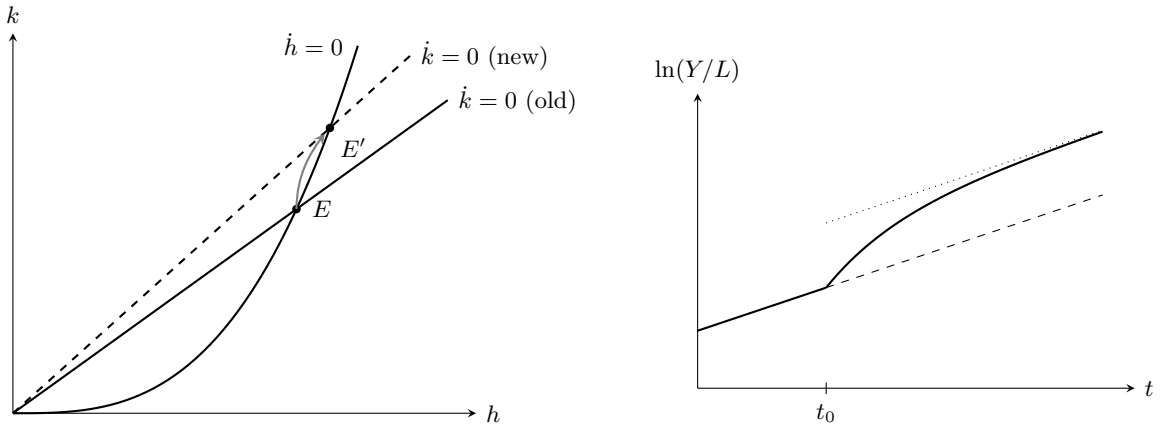


Figure 4.8-2. Left: the rise in s steepens the $\dot{k}=0$ ray; the economy moves from E to E' inside the new region III. Right: $\ln(Y/L)$ (solid) does not jump at t_0, grows faster than g during the transition, and approaches a path parallel to the old one (dashed) but higher (dotted).

Economic intuition. A higher s raises K, which raises goods output directly, as in Solow, and also raises the inputs to schooling and hence H: a second round of accumulation that Solow lacks. That is why the long-run elasticity exceeds the Solow value $\alpha/(1-\alpha)$ by $\gamma/[(1-\alpha)(1-\gamma-\phi)]$. Neither round changes long-run growth, which is pinned down by g: a level effect, not a growth effect.

Where this goes wrong. Two common slips: letting output per person jump at t_0 (only its growth rate jumps, because k and h are stocks), and having the economy reach E' at a finite date (convergence is asymptotic).

Ref: Romer (2012), §4.1, pp. 151–156 (human capital in the Solow framework; the intensive-form method of eq. (4.7)); §1.3–1.5, pp. 15–26 (Solow dynamics and the level effect of s); Problem 4.9 and Lucas (1988) for the endogenous-growth case.

Checked: `check-4-8-8-1.py` verifies the laws of motion from the levels, both loci and the signs off them, k^* and h^*, the Jacobian's trace, determinant and discriminant, growth at g on the balanced path (simulated in levels), the jump and initial slope of growth after a rise in s, the s-elasticity, monotone convergence to E' and convergence from 20 random starting points. The solution manual agrees on (a)–(d); its sketch has the economy reach E' at a finite date t_1, whereas convergence is only asymptotic.

Romer 8.1 — Life-cycle saving

Statement in the book, p. 398. Background: Romer §8.1.

Notation. $Y(t)$ is income, $W(t)$ wealth (net assets, possibly negative), $S(t)\equiv\dot{W}(t)$ saving and $\bar{C}$ the optimal consumption level. I take $Y_0>0$ and $g\geq 0$ (income rising with experience); the book does not sign g, and only the location of the peak of wealth in part (c) depends on it.

(a)

Step — flow constraint. With a zero interest rate, wealth earns nothing and changes only through saving:

$$\dot{W}(t) = Y(t) - C(t), \qquad W(0) = 0.$$

The individual may borrow but must die without debt, $W(T) \geq 0$: the continuous-time version of the requirement behind Romer's eq. (8.2) that any debt be repaid by the end of life.

Step — integrate. Integrating from 0 to T, $W(T) = W(0) + \int_0^T [Y(t) - C(t)]\,dt \geq 0$, where lifetime income is

$$\int_0^T Y(t)\,dt = \int_0^R (Y_0 + gt)\,dt + \int_R^T 0\,dt = \Big[Y_0 t + \frac{g}{2}t^2\Big]_0^R = Y_0 R + \frac{g}{2}R^2.$$

So the lifetime budget constraint is

$$\int_0^T C(t)\,dt \;\leq\; Y_0 R + \frac{g}{2}R^2 = R\Big(Y_0 + \frac{gR}{2}\Big).$$

With $r = 0$ present values are plain sums: lifetime resources are the working span R times average income over the working life, $Y_0 + gR/2$ (income at mid-career).

(b)

Step — the constraint binds. If it were slack, raising C on a short interval would keep the path feasible and, since $u' > 0$, raise U. So the constraint holds with equality.

Step — first-order condition. With multiplier λ (Romer's eq. (8.3) in continuous time),

$$\begin{aligned}\mathcal{L} &= \int_0^T u(C(t))\,dt + \lambda\Big[Y_0 R + \frac{g}{2}R^2 - \int_0^T C(t)\,dt\Big] \\ &= \int_0^T \big[u(C(t)) - \lambda C(t)\big]\,dt + \lambda\Big(Y_0 R + \frac{g}{2}R^2\Big).\end{aligned}$$

Consumption at date t enters only the integrand at t, so $\mathcal{L}$ is maximized date by date: $u'(C(t)) = \lambda$ for every $t \in [0, T]$, retirement included. As $u'' < 0$, u' is strictly decreasing, hence one-to-one, so $C(t) = (u')^{-1}(\lambda) \equiv \bar{C}$ for all t: consumption is flat.

Step — the level. Substituting the flat path into the binding constraint, $\bar{C}\,T = Y_0 R + \frac{g}{2}R^2$, so

$$C(t) = \bar{C} = \frac{R}{T}\Big(Y_0 + \frac{gR}{2}\Big) = \frac{1}{T}\int_0^T Y(\tau)\,d\tau \qquad \text{for all } t \in [0, T].$$

Step — sufficiency. For any path that satisfies the constraint, Jensen's inequality for the strictly concave u, and then $u' > 0$, give

$$\frac{1}{T}\int_0^T u(C(t))\,dt \;\leq\; u\Big(\frac{1}{T}\int_0^T C(t)\,dt\Big) \;\leq\; u(\bar{C}),$$

with equality only if $C(t) = \bar{C}$ (almost everywhere). So the flat path is the unique optimum, not merely a stationary point.

Economic intuition. With zero interest and no discounting, a unit of consumption costs the same and is valued through the same u at every date, so diminishing marginal utility makes the individual equalize consumption across dates. $\bar{C}$ is permanent income, Romer's eq. (8.5): it depends on lifetime resources, not on when income arrives, and it is the same in retirement, when income is zero.

(c)

Step — saving. $S(t) = Y(t) - \bar{C}$: $S(t) = Y_0 + gt - \bar{C}$ for $0 \leq t < R$, and $S(t) = -\bar{C} < 0$ for $R \leq t \leq T$.

Step — working years, $0 \leq t \leq R$. Integrating saving from $W(0) = 0$, and then using $Y_0 R + \frac{g}{2}R^2 = \bar{C}T$ at $t = R$,

$$W(t) = \int_0^t (Y_0 + g\tau - \bar{C})\,d\tau = (Y_0 - \bar{C})\,t + \frac{g}{2}t^2, \qquad W(R) = Y_0 R + \frac{g}{2}R^2 - \bar{C}R = \bar{C}\,(T - R).$$

Step — retirement, $R \le t \le T$**.** Wealth falls at the constant rate $\bar{C}$ from $W(R)$:

$$W(t) = W(R) - \int_R^t \bar{C}\, d\tau = \bar{C}\,(T-R) - \bar{C}\,(t-R) = \bar{C}\,(T-t),$$

so $W(T) = 0$: the budget is exhausted exactly at death, and in retirement wealth always equals the consumption still to be financed.

Step — the hump. On $[0, R)$, $\ddot{W} = g \ge 0$, so W is convex there and its maximum over $[0, R]$ is at an endpoint, where $W(R) = \bar{C}(T-R) > 0 = W(0)$. Saving just before retirement is positive,

$$S(R^-) = Y_0 + gR - \bar{C} \;\ge\; Y_0 + \frac{gR}{2} - \bar{C} = \Big(1 - \frac{R}{T}\Big)\Big(Y_0 + \frac{gR}{2}\Big) > 0,$$

and on $(R, T]$ wealth falls at rate $\bar{C}$. So wealth peaks at retirement, with a kink there (saving drops from $S(R^-) > 0$ to $-\bar{C}$):

$$\max_{0 \le t \le T} W(t) = W(R) = \frac{R\,(T-R)}{T}\Big(Y_0 + \frac{gR}{2}\Big).$$

Step — borrowing when young. Initial saving $S(0) = Y_0 - \bar{C}$ is negative iff $Y_0 T < RY_0 + \frac{g}{2}R^2$, i.e. iff $gR^2 > 2Y_0(T-R)$: steep income growth. Wealth then falls until income reaches $\bar{C}$ at $t_m = (\bar{C} - Y_0)/g$, where $W(t_m) = -(\bar{C}-Y_0)^2/g + (\bar{C}-Y_0)^2/(2g) = -(\bar{C}-Y_0)^2/(2g) < 0$. Since $W(t) = t\,[\,Y_0 - \bar{C} + \frac{g}{2}t\,]$, it is back to zero at $\hat{t} = 2(\bar{C}-Y_0)/g$, which precedes R because $\bar{C} < Y_0 + gR/2$, and it then climbs to the peak at R. If $Y_0 \ge \bar{C}$, wealth is never negative.

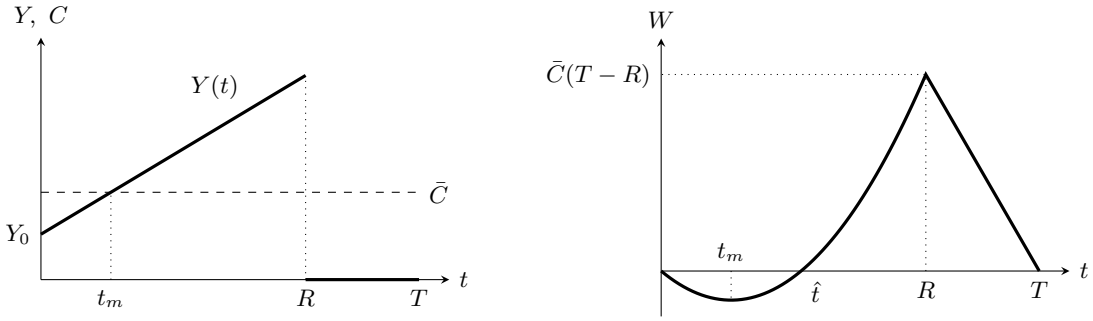


Figure 8.1-1. Left: income (thick) and consumption (dashed). Right: wealth. Drawn for $Y_0 = 0.6$, $g = 0.3$, $R = 7$, $T = 10$, so $\bar{C} = 1.155$ and $gR^2 > 2Y_0(T-R)$: the young borrow until $\hat{t} = 3.7$, and wealth peaks at $W(R) = 3.465$.

Economic intuition. Saving is future consumption: the individual saves when income is above its lifetime average and dissaves when it is below, so wealth is the running area between the income line and the flat consumption line. It is built up over the working life and run down to exactly zero in retirement: the hump of Modigliani and Brumberg.

Where this goes wrong. The peak at R uses $Y(R^-) \ge \bar{C}$, which $g \ge 0$ guarantees. If $g < 0$ and income falls to $\bar{C}$ before retirement, at $t_p = (Y_0 - \bar{C})/|g| < R$, wealth is concave on $[0, R]$ and peaks at t_p, while $W(T) = 0$ still holds. Nor should $W(t) \ge 0$ be imposed: the book allows borrowing, and it is optimal when income grows steeply.

Ref: Romer (2012), §8.1, pp. 365–368, eqs. (8.1)–(8.6) (the discrete-time version: flat consumption, permanent income, saving as income minus its lifetime average); Modigliani and Brumberg (1954).

Checked: `check-4-8-8-1.py` verifies with sympy the lifetime resources, $\bar{C}$, both pieces of $W(t)$, $W(R) = \bar{C}(T-R)$, $W(T) = 0$, the borrowing condition, t_m, $W(t_m)$ and $\hat{t} < R$; numerically, that W peaks at R for random (Y_0, g, R, T) with $g \ge 0$, that budget-neutral perturbations of the flat path lower utility (log and CRRA u), and the values plotted in Figure 8.1-1. The solution manual agrees; it draws only the case $Y_0 > \bar{C}$, without early borrowing. **Romer**

8.13 — Habit formation

Statement in the book, pp. 402–403. Background: Romer §8.6.

Step — Notation. $\beta \equiv 1/(1+\rho)$ and $u(C,Z) \equiv (C/Z)^{1-\theta}/(1-\theta)$, so lifetime utility is $U_i = \sum_{t=1}^{T} \beta^t u(C_{it}, Z_{it})$; the interest rate r is constant and there is no uncertainty. The derivatives used throughout are

$$u_C(C,Z) = \frac{1}{Z}\Big(\frac{C}{Z}\Big)^{-\theta} = C^{-\theta}Z^{\theta-1}, \qquad u_Z(C,Z) = -\frac{C}{Z^2}\Big(\frac{C}{Z}\Big)^{-\theta} = -C^{1-\theta}Z^{\theta-2},$$
$$u_{CZ}(C,Z) = (\theta-1)\,C^{-\theta}Z^{\theta-2}, \quad \text{positive exactly when } \theta > 1.$$

(a)

Step — The perturbation, (i). Lower C_{it} by dC and raise $C_{i,t+1}$ by $(1+r)\,dC$. In the present-value budget, $\sum_t C_{it}/(1+r)^t \le A_{i0} + \sum_t Y_{it}/(1+r)^t$, spending changes by $-dC/(1+r)^t + (1+r)\,dC/(1+r)^{t+1} = 0$: the change is feasible in either direction, so at an optimum its first-order effect on U_i is zero. The reference levels $Z_{it} = C_{t-1}^{\phi}$ and $Z_{i,t+1} = C_t^{\phi}$ are aggregates that i takes as given, so only the terms of t and $t+1$ move:

$$dU_i = -\beta^t C_{it}^{-\theta} Z_{it}^{\theta-1}\,dC + \beta^{t+1} C_{i,t+1}^{-\theta} Z_{i,t+1}^{\theta-1}(1+r)\,dC = 0$$
$$\Longrightarrow \quad C_{it}^{-\theta} Z_{it}^{\theta-1} = \frac{1+r}{1+\rho}\, C_{i,t+1}^{-\theta} Z_{i,t+1}^{\theta-1} \qquad \text{(dividing by } \beta^t dC\text{)}.$$

Step — Euler equation, (i). Substitute $Z_{it} = C_{t-1}^{\phi}$ and $Z_{i,t+1} = C_t^{\phi}$, then divide both sides by $C_{i,t+1}^{-\theta} C_{t-1}^{\phi(\theta-1)}$:

$$\Big(\frac{C_{i,t+1}}{C_{it}}\Big)^{\theta} = \frac{1+r}{1+\rho}\Big(\frac{C_t}{C_{t-1}}\Big)^{\phi(\theta-1)} \quad \Longrightarrow \quad \frac{C_{i,t+1}}{C_{it}} = \Big(\frac{1+r}{1+\rho}\Big)^{1/\theta}\Big(\frac{C_t}{C_{t-1}}\Big)^{\phi(\theta-1)/\theta}.$$

For $\phi = 0$ this is the standard CRRA Euler equation, Romer's (8.26).

Step — Equilibrium, (ii). The representative consumer's consumption is aggregate consumption, $C_{it} = C_t$ for all t, so the left side becomes C_{t+1}/C_t. Taking logs (exact, no approximation):

$$\ln C_{t+1} - \ln C_t = a + b\,(\ln C_t - \ln C_{t-1}), \qquad a \equiv \frac{\ln(1+r) - \ln(1+\rho)}{\theta} \simeq \frac{r-\rho}{\theta}, \qquad b \equiv \frac{\phi(\theta-1)}{\theta}.$$

Step — $\phi > 0$ and $\theta = 1$. Then $b = 0$ and $\ln C_{t+1} - \ln C_t = \ln(1+r) - \ln(1+\rho)$ every period: exactly the path without habits, so habit formation does not affect behavior. At $\theta = 1$ period utility is $\ln C_{it} - \phi \ln C_{t-1}$ and $u_C = 1/C_{it}$ does not depend on Z_{it}: the habit subtracts a term the individual cannot influence, which lowers the level of utility but changes no marginal rate of substitution.

Step — $\phi > 0$ and $\theta > 1$. Then $0 < b \le (\theta-1)/\theta < 1$ (because $\phi \le 1$): consumption growth is positively serially correlated. With $x_t \equiv \ln C_t - \ln C_{t-1}$, the equation $x_{t+1} = a + b x_t$ has the fixed point and dynamics

$$x^* = \frac{a}{1-b} = \frac{\ln(1+r) - \ln(1+\rho)}{\theta - \phi(\theta-1)}, \qquad x_{t+1} - x^* = b\,(x_t - x^*),$$

so a burst of growth dies out only gradually, at rate b per period. Since $1 \le \theta - \phi(\theta-1) < \theta$, long-run growth also responds more to $r-\rho$ than without habits. (For $\theta < 1$, $b < 0$ and growth rates alternate around x^*.)

Economic intuition. For $\theta > 1$, $u_{CZ} > 0$: a higher reference level raises the marginal utility of one's own consumption ("catching up with the Joneses"). If aggregate consumption has just grown, the reference level is higher at $t+1$ than at t, so marginal utility at $t+1$ is high relative to t and each consumer tilts spending toward $t+1$, i.e. grows faster. Everyone does the same, so past aggregate growth feeds current growth. With $\theta = 1$ the cross effect u_{CZ} is zero and the habit is behaviorally irrelevant.

(b)

Step — Where C_{it} enters, (i). Now $Z_{it} = C_{i,t-1}$ and $\phi = 1$: C_{it} is the numerator of the period-t term and the reference level of the period-$(t+1)$ term. Write $u_C^{(s)} \equiv u_C(C_{is}, C_{i,s-1})$ and $u_Z^{(s)} \equiv u_Z(C_{is}, C_{i,s-1})$. The perturbation of (a)(i) moves C_{it} by $-dC$ and $C_{i,t+1}$ by $(1+r)\,dC$, so it changes utility at t, $t+1$ and $t+2$:

$$\begin{aligned} dU_i &= \beta^t\big[-u_C^{(t)}\big]dC + \beta^{t+1}\big[(1+r)\,u_C^{(t+1)} - u_Z^{(t+1)}\big]dC + \beta^{t+2}(1+r)\,u_Z^{(t+2)}\,dC \\ &= -\beta^t M_t\,dC + \beta^{t+1}(1+r)\,M_{t+1}\,dC, \qquad M_s \equiv u_C^{(s)} + \beta\,u_Z^{(s+1)}. \end{aligned}$$

M_s is the full marginal utility of C_{is} in date-s units; its second term, negative, is the habit cost (consuming more at s raises the reference level at $s+1$). At T there is no such cost: $M_T = u_C^{(T)}$.

Step — Euler equation, (i). Setting $dU_i = 0$ gives $M_t = \frac{1+r}{1+\rho}M_{t+1}$, i.e. with u_C, u_Z from the notation step,

$$C_{it}^{-\theta}C_{i,t-1}^{\theta-1} - \frac{C_{i,t+1}^{1-\theta}C_{it}^{\theta-2}}{1+\rho} = \frac{1+r}{1+\rho}\left[C_{i,t+1}^{-\theta}C_{it}^{\theta-1} - \frac{C_{i,t+2}^{1-\theta}C_{i,t+1}^{\theta-2}}{1+\rho}\right], \qquad t \le T-2$$

(for $t = T-1$ the last term is absent). Let $X_s \equiv C_{is}/C_{i,s-1} = 1+g_s$. Since $C_{is}^{-\theta}C_{i,s-1}^{\theta-1} = X_s^{1-\theta}/C_{is}$ and $C_{i,s+1}^{1-\theta}C_{is}^{\theta-2} = X_{s+1}^{1-\theta}/C_{is}$, multiplying by $C_{i,t+1}$ and using $C_{i,t+1}/C_{it} = X_{t+1}$ gives the growth form

$$X_{t+1}\Big[X_t^{1-\theta} - \frac{X_{t+1}^{1-\theta}}{1+\rho}\Big] = \frac{1+r}{1+\rho}\Big[X_{t+1}^{1-\theta} - \frac{X_{t+2}^{1-\theta}}{1+\rho}\Big].$$

Only growth rates appear because u is homogeneous of degree zero in $(C_{it}, C_{i,t-1})$.

Step — Linearization, (ii). With $\rho = r = 0$ the growth form is $X_{t+1}[X_t^{1-\theta} - X_{t+1}^{1-\theta}] = X_{t+1}^{1-\theta} - X_{t+2}^{1-\theta}$. Replace each power by $X_s^{1-\theta} \simeq 1 + (1-\theta)g_s$ (the suggested approximation with $\gamma = 1-\theta$) and X_{t+1} by $1+g_{t+1}$:

$$\text{left side:}\quad (1+g_{t+1})(1-\theta)(g_t - g_{t+1}) = (1-\theta)(g_t - g_{t+1}) + \underbrace{(1-\theta)\,g_{t+1}(g_t - g_{t+1})}_{\text{interaction: dropped}},$$

$$\text{right side:}\quad (1-\theta)(g_{t+1} - g_{t+2}).$$

Equating the two and dividing by $-(1-\theta)$, which requires $\theta \neq 1$:

$$g_{t+2} - g_{t+1} = g_{t+1} - g_t.$$

Keeping the interaction term instead gives $g_{t+2} - g_{t+1} = (1+g_{t+1})(g_{t+1} - g_t)$.

Step — Direction of the drift (exact). With $\rho = r = 0$ the Euler equations give $M_1 = M_2 = \cdots = M_T \equiv \mu$, and $\mu = M_T = X_T^{1-\theta}/C_{iT} > 0$. With $m_s \equiv X_s^{1-\theta}$ and $m_{T+1} \equiv 0$, $M_s = (m_s - m_{s+1})/C_{is}$ for every $s \le T$, so $m_s - m_{s+1} = \mu C_{is}$; summing over $s = t, \ldots, T$,

$$\Big(\frac{C_{it}}{C_{i,t-1}}\Big)^{1-\theta} = \mu\sum_{s=t}^{T} C_{is} \equiv \mu R_t,$$

where R_t is what remains to be consumed from t on. For $\theta > 1$, $1 + g_t = (\mu R_t)^{-1/(\theta-1)}$ rises as R_t falls, and to first order the constant step is $g_{t+1} - g_t \simeq \mu C_{it}/(\theta-1) > 0$. Example ($T = 6$, $\theta = 3$, $C_{i0} = 1$, $\sum_t C_{it} = 6$): consumption is U-shaped and $1 + g_t$ rises from 0.85 to 1.60.

Economic intuition. Internal habits make period utility a function of growth alone, so the consumer smooths growth the way a standard consumer smooths levels. But resources move in levels: shifting a small amount from t to $t+1$ changes $(\ln X_t, \ln X_{t+1}, \ln X_{t+2})$ by $(-\epsilon, +2\epsilon, -\epsilon)$, and $\partial u/\partial \ln X = X^{1-\theta}$, so optimality sets the second difference of $X^{1-\theta}$, hence (to first order) of g, to zero. Growth then changes by the same amount every period: an acceleration persists one for one, far more persistence than external habits deliver ($b < 1$). The sign comes from the end of life: the last period carries no

habit cost, so consumption is back-loaded and its growth speeds up (in the example it first falls, lowering the habit stock).

Where this goes wrong. Part (b) is not part (a) with $\phi = 1$: there $Z_{i,t+1}$ is taken as given, here the u_Z term is internalized. And the division by $1-\theta$ is not innocuous. At $\theta = 1$ with $\rho = 0$ lifetime utility telescopes to $\ln C_{iT} - \ln C_{i0}$ and the Euler equation reads $0 = 0$; for $\theta < 1$ letting $C_{it} \to 0$ for one period makes $(C_{i,t+1}/C_{it})^{1-\theta}$, and so U_i, unbounded. With $\rho = 0$, as in (ii), the problem has a maximum only for $\theta > 1$.

Ref: Romer §8.4, pp. 380–381, eqs. (8.24)–(8.26) (CRRA Euler equation by the same perturbation); §8.6, p. 390, n. 14 (habit-formation literature); ch. 7, p. 358 (habits as a source of sluggish consumption).

Checked: `check-8-13-8-15.py` differentiates the perturbed utility symbolically and confirms, at random points, the (a)(i) Euler ratio and the (b)(i) Euler equation and its growth form; checks $b = 0$ at $\theta = 1$, $0 < b < 1$ for $\theta > 1$ and x^*; series-expands the (b) condition to first order; and solves the $T = 6$ example numerically, confirming M_t constant, $X_t^{1-\theta}/R_t$ constant and rising growth. The Romer solution manual agrees with (a) and (b); it asserts without proof that growth increases over time, which the exact step establishes for $\theta > 1$.

Romer 8.15 — Time-inconsistent preferences

Statement in the book, pp. 403–404. Background: Romer §8.6.

Step — Environment. With $r = 0$ the only constraint is $c_1 + c_2 + c_3 = W$. The date-1 self weights $(\ln c_1, \ln c_2, \ln c_3)$ by $(1, \delta, \delta)$ and the date-2 self weights $(\ln c_2, \ln c_3)$ by $(1, \delta)$: quasi-hyperbolic (β–δ) discounting with short-run factor δ and long-run factor 1. Every future date is discounted by δ relative to "now", but two future dates are not discounted relative to each other. Every objective below is a sum of strictly concave functions of the choice variables on a convex set, so first-order conditions identify the maxima.

(a)

Step — (i) Commitment. Substitute $c_3 = W - c_1 - c_2$ into the date-1 objective and differentiate with respect to c_1 and c_2:

$$\frac{1}{c_1} - \frac{\delta}{W - c_1 - c_2} = 0, \qquad \frac{\delta}{c_2} - \frac{\delta}{W - c_1 - c_2} = 0.$$

The second condition gives $c_2 = c_3$; the first gives $c_3 = \delta c_1$. The budget then reads $c_1 + 2\delta c_1 = W$:

$$c_1^C = \frac{W}{1+2\delta}, \qquad c_2^C = c_3^C = \frac{\delta W}{1+2\delta}.$$

Step — The date-2 self's rule. At date 2 the remaining wealth is $W_2 \equiv W - c_1$ and the date-2 self maximizes $\ln c_2 + \delta \ln(W_2 - c_2)$: $1/c_2 = \delta/(W_2 - c_2)$, so $W_2 - c_2 = \delta c_2$ and

$$c_2 = \frac{W_2}{1+\delta}, \qquad c_3 = \frac{\delta W_2}{1+\delta}.$$

The rule is proportional to wealth, with date-2 share $s^* \equiv 1/(1+\delta) > 1/2$.

Step — (ii) Naivete. The naif expects to follow the date-1 plan, so it solves the commitment problem and picks $c_1^N = c_1^C$. Then $W_2 = W - c_1^N = 2\delta W/(1+2\delta)$ and the date-2 rule gives

$$c_1^N = \frac{W}{1+2\delta}, \qquad c_2^N = \frac{2\delta}{1+\delta} \cdot \frac{W}{1+2\delta}, \qquad c_3^N = \frac{2\delta^2}{1+\delta} \cdot \frac{W}{1+2\delta}.$$

Step — (iii) Sophistication. The sophisticate substitutes the date-2 rule, with $W_2 = W - c_1$, into its date-1 objective:

$$\ln c_1 + \delta \ln \frac{W - c_1}{1+\delta} + \delta \ln \frac{\delta(W - c_1)}{1+\delta} = \ln c_1 + 2\delta \ln(W - c_1) + \delta \ln \frac{\delta}{(1+\delta)^2}.$$

The first-order condition $1/c_1 = 2\delta/(W - c_1)$ gives $W - c_1 = 2\delta c_1$, so $c_1^S = W/(1+2\delta) = c_1^N$; the same date-2 rule then gives $c_2^S = c_2^N$ and $c_3^S = c_3^N$.

(b)

Step — (i) In what sense preferences are time-inconsistent. Seen from date 1, c_2 and c_3 carry the same weight, so the plan is $c_3/c_2 = 1$. At date 2 the same person, with no new information, sets $c_3/c_2 = \delta < 1$; comparing (a)(ii) with (a)(i),

$$\frac{c_2^N}{c_2^C} = \frac{2}{1+\delta} > 1, \qquad \frac{c_3^N}{c_3^C} = \frac{2\delta}{1+\delta} < 1.$$

The marginal rate of substitution between c_2 and c_3 depends on the date at which it is evaluated, not only on the dates at which consumption occurs, so the plan that is optimal at date 1 is no longer optimal at date 2 and is abandoned. (With exponential weights $(1, \delta, \delta^2)$ and $(1, \delta)$ the ratio of the weights on c_3 and c_2 is δ at both dates, and plans are kept.) The date-1 self loses from the revision, since c_1 is the same in both cases:

$$U_1^C - U_1^N = \delta \ln \frac{c_2^C c_3^C}{c_2^N c_3^N} = \delta \ln \frac{(1+\delta)^2}{4\delta} > 0,$$

because $(1+\delta)^2 - 4\delta = (1-\delta)^2 > 0$; it would pay for a commitment device, such as the illiquid saving Romer mentions on p. 398.

Step — (ii) Why sophistication changes nothing. Let the date-2 split be any rule proportional to wealth, $c_2 = sW_2$ and $c_3 = (1-s)W_2$, with $s \in (0,1)$ independent of W_2. The date-1 objective is then

$$\ln c_1 + \delta \ln[s(W - c_1)] + \delta \ln[(1-s)(W - c_1)] = \ln c_1 + 2\delta \ln(W - c_1) + \delta \ln[s(1-s)],$$

so s enters only through an additive constant, and the first-order condition $1/c_1 = 2\delta/(W - c_1)$ does not contain s: $c_1 = W/(1+2\delta)$ for every s. The naif uses $s = 1/2$, the sophisticate $s = s^*$; both choose the same c_1, after which the same date-2 self chooses c_2 and c_3. Equivalently, the marginal value of saving at date 1 is

$$\delta\Big[\frac{1}{c_2}\frac{\partial c_2}{\partial W_2} + \frac{1}{c_3}\frac{\partial c_3}{\partial W_2}\Big] = \delta\Big[\frac{s}{sW_2} + \frac{1-s}{(1-s)W_2}\Big] = \frac{2\delta}{W_2},$$

independent of s.

Economic intuition. The sophisticate knows that a unit saved will go mostly to c_2, where its marginal utility is low, and less to c_3, where it is high. With log utility, marginal utility times consumption is one, so each share times its marginal utility equals $1/W_2$ and the two effects cancel exactly. Put differently, the future self's misallocation works like a lower return on saving, and with log utility the income and substitution effects of a change in the return offset each other.

Step — What changes with CRRA utility. Replace $\ln c$ by $u(c) = c^{1-\sigma}/(1-\sigma)$, $\sigma > 0$, $\sigma \neq 1$, in both objectives. Commitment: $c_1^{-\sigma} = \delta c_2^{-\sigma} = \delta c_3^{-\sigma}$, so $c_2 = c_3 = \delta^{1/\sigma} c_1$ and $c_1^C = c_1^N = W/(1+2\delta^{1/\sigma})$. Date 2: $c_2^{-\sigma} = \delta c_3^{-\sigma}$ gives $c_3 = \delta^{1/\sigma} c_2$, again a proportional rule, with $s^* = 1/(1+\delta^{1/\sigma})$. For any share s, $u(sW_2) = s^{1-\sigma} u(W_2)$, so the date-1 problem and its first-order condition are

$$\max_{c_1} \; u(c_1) + \delta A(s)\, u(W - c_1), \qquad A(s) \equiv s^{1-\sigma} + (1-s)^{1-\sigma},$$

$$c_1^{-\sigma} = \delta A(s)\,(W - c_1)^{-\sigma} \implies c_1 = \frac{W}{1 + [\delta A(s)]^{1/\sigma}}.$$

Now s scales the marginal value of saving instead of shifting a constant. The naif's $s = 1/2$ gives $A = 2^\sigma$ and returns the commitment c_1. A is symmetric about $1/2$ with $A''(s) = -\sigma(1-\sigma)[s^{-\sigma-1} + (1-s)^{-\sigma-1}]$, and $s^* \neq 1/2$. For $\sigma < 1$, A is concave with its maximum at $1/2$, so $A(s^*) < 2^\sigma$ and $c_1^S > c_1^N$: the substitution effect dominates and the sophisticate saves less. For $\sigma > 1$, A is convex

with its minimum at 1/2, so $c_1^S < c_1^N$: the income effect dominates and it saves more to offset its future self's overspending. With $\delta = 1/2$: $\sigma = 2$ gives $c_1^S = 0.4106W < c_1^N = 0.4142W$, and $\sigma = 1/2$ gives $c_1^S = 0.6897W > c_1^N = 0.6667W$.

Where this goes wrong. Naivete and commitment share c_1 by construction, so the inconsistency shows up only in (c_2, c_3); comparing first-period choices cannot reveal it. And the equivalence of naive and sophisticated choices is a property of log utility, not of the β–δ structure: the date-2 rule is proportional to wealth for every CRRA utility, but only under log does the share drop out of the marginal value of saving.

Ref: Romer §8.6, pp. 397–398 (time-inconsistent preferences; Laibson, 1997); §8.4, pp. 381–383 (income and substitution effects of the interest rate).

Checked: `check-8-13-8-15.py` solves the commitment, date-2, naive and sophisticated problems symbolically, verifies that the cross derivative $\partial^2/\partial c_1 \partial s$ of the date-1 objective under an arbitrary share s is zero, checks $U_1^C - U_1^N = \delta \ln[(1+\delta)^2/(4\delta)] > 0$, and for $\sigma \in \{2, 1/2\}$ re-derives s^*, c_1^N and c_1^S by numerical maximization, matching the closed forms and the printed numbers. The Romer solution manual agrees; its (b)(ii) gives the same two offsetting effects and states, without derivation, that with other utility functions either ordering is possible, which the CRRA step makes precise.

Romer 10.16 — Search efficiency with a general matching function

Statement in the book, p. 512. Background: Romer §10.7.

Step — Environment. The static economy of eqs. (10.71)–(10.75): a given pool of U unemployed workers; V vacancies, each created at cost c; $E = M(U,V)$ matches, each producing y; wage $w = \phi y$; no income for the unemployed. The wage is a transfer between matched parties, so welfare is $W(V) = M(U,V)\, y - cV$. Only the matching function changes. "Well-behaved" is taken to mean: M is smooth, $M_V > 0$, $M_{VV} < 0$, $M(U,0) = 0$ and $M \le V$ (a filling probability cannot exceed one); neither Cobb–Douglas form nor constant returns is assumed. Because U is fixed, the only margin is V. Define the job-filling probability q and the elasticity of matches with respect to vacancies η:

$$q(V) \equiv \frac{M(U,V)}{V}, \qquad \eta(V) \equiv \frac{V\, M_V(U,V)}{M(U,V)}.$$

Step — The elasticity lies in $(0,1)$. Strict concavity puts the tangent line at V above the function at 0: $M(U,0) < M(U,V) + M_V(U,V)\,(0 - V)$. With $M(U,0) = 0$ this reads $VM_V < M$, so $\eta(V) < 1$; and $M_V > 0$ gives $\eta(V) > 0$. Differentiating q,

$$q'(V) = \frac{VM_V - M}{V^2} = -\frac{q(V)}{V}\,[1 - \eta(V)] < 0.$$

Step — Decentralized equilibrium. A vacancy is filled with probability $q(V)$, and a filled job leaves the firm $y - w = (1-\phi)y$. The expected value of opening a vacancy is $q(V)(1-\phi)y - c$; free entry drives it to zero (eq. (10.71) with a general M):

$$\frac{M(U, V^{EQ})}{V^{EQ}}\,(1-\phi)\, y = c. \qquad \text{(F)}$$

The left side is strictly decreasing in V because $q' < 0$, so (F) has at most one root; it exists ($V^{EQ} > 0$) by assumption, hence it is unique.

Step — Planner. The planner chooses V to maximize $W(V) = M(U,V)\, y - cV$:

$$W'(V) = M_V(U,V)\, y - c, \qquad W''(V) = M_{VV}(U,V)\, y < 0.$$

W is strictly concave, so the first-order condition (eq. (10.74) with a general M)

$$M_V(U, V^*)\, y = c \qquad \text{(P)}$$

is necessary and sufficient, and V^* is unique.

Step — The wedge between (P) and (F). By the definition of η, $M_V = \eta(V)\, M/V = \eta(V)\, q(V)$ at every V, so $W'(V) = \eta(V)\, q(V)\, y - c$. At $V = V^{EQ}$, replace c using (F):

$$W'(V^{EQ}) = \eta(V^{EQ})\, q(V^{EQ})\, y - (1-\phi)\, q(V^{EQ})\, y = q(V^{EQ})\, y\, [\eta(V^{EQ}) - (1-\phi)].$$

Step — Efficiency condition. ($\Leftarrow$) If $\eta(V^{EQ}) = 1-\phi$, the bracket is zero, so $W'(V^{EQ}) = 0$: V^{EQ} solves (P), whose only root is V^*; hence $V^{EQ} = V^*$. ($\Rightarrow$) If $V^{EQ} = V^*$, then $W'(V^{EQ}) = 0$; since $q(V^{EQ})\, y > 0$, the bracket vanishes: $\eta(V^{EQ}) = 1-\phi$. Therefore

$$V^{EQ} = V^* \quad \Longleftrightarrow \quad \left.\frac{V\, M_V(U,V)}{M(U,V)}\right|_{V=V^{EQ}} = 1-\phi.$$

The answer is **yes**: the condition survives without Cobb–Douglas form or constant returns, with one qualification — the elasticity is evaluated at the equilibrium allocation. Evaluating it at V^* instead is equivalent: if $\eta(V^*) = 1-\phi$, then by (P) $q(V^*)(1-\phi)y = \eta(V^*)\, q(V^*)\, y = M_V(U,V^*)\, y = c$, so V^* solves (F), and since the root of (F) is unique, $V^{EQ} = V^*$.

Step — Direction of the inefficiency. W' is strictly decreasing and $W'(V^*) = 0$. If $\eta(V^{EQ}) < 1-\phi$, then $W'(V^{EQ}) < 0$, so $V^{EQ} > V^*$: too many vacancies. If $\eta(V^{EQ}) > 1-\phi$, then $W'(V^{EQ}) > 0$ and $V^{EQ} < V^*$: too few. This is Romer's result for $\gamma \lessgtr 1-\phi$, with γ replaced by the elasticity at the equilibrium point.

Step — What a general M changes. Under Cobb–Douglas, $\eta \equiv \gamma$ at every point, so the condition restricts parameters only. In general η varies with (U,V) and V^{EQ} depends on ϕ, y, c and U, so the condition restricts the equilibrium point. Example, neither Cobb–Douglas nor constant-returns: $M = UV/(1+U+V)$, which meets every assumption above. Here $M_V = U(1+U)/(1+U+V)^2$, $q = U/(1+U+V)$ and $\eta = (1+U)/(1+U+V)$, so

$$\text{(F)}:\ 1+U+V^{EQ} = \frac{(1-\phi)\, yU}{c}, \qquad \text{(P)}:\ \left(1+U+V^*\right)^2 = \frac{yU(1+U)}{c},$$
$$V^{EQ} = V^* \iff 1-\phi = \Big[\frac{c\,(1+U)}{yU}\Big]^{1/2}.$$

The last equivalence squares (F) and compares it with (P). With $U=1$, $y=8$, $c=1$: $\phi = \frac{1}{2}$ gives $V^{EQ} = V^* = 2$ and $\eta(V^{EQ}) = \frac{1}{2} = 1-\phi$; $\phi = \frac{1}{4}$ gives $V^{EQ} = 4 > V^* = 2$, with $\eta(V^{EQ}) = \frac{1}{3} < \frac{3}{4}$. The efficient share moves with c/y and with the size of the unemployment pool, so no fixed bargaining rule is efficient for all fundamentals.

Economic intuition. Split the gap between the social and the private value of a marginal vacancy:

$$\underbrace{M_V\, y}_{\text{social}} - \underbrace{q\,(1-\phi)\, y}_{\text{private}} = \underbrace{\phi\, y\, M_V}_{\text{gain to workers}} - \underbrace{(1-\phi)\, y\, (q - M_V)}_{\text{congestion on other firms}}.$$

The entrant expects q matches, but total matches rise only by $M_V = \eta q$: the other $q - M_V = (1-\eta)q$ are taken from other vacancies, each worth $(1-\phi)y$ to its owner, while each new match hands ϕy to a worker. The two externalities cancel iff $M_V = (1-\phi)q$, that is, $\eta = 1-\phi$. The accounting never uses the functional form; it only has to balance where entry stops.

Where this goes wrong. (i) Treating $\eta = 1-\phi$ as a restriction on parameters: with a general M it must hold at V^{EQ}, which itself moves with ϕ; a point $\hat{V} \neq V^{EQ}$ with $\eta(\hat{V}) = 1-\phi$ proves nothing. (ii) Thinking constant returns are needed: U is fixed here, so only the V-elasticity enters; in the dynamic model constant returns matter only because they make η a function of tightness V/U alone.

Ref: Romer (2012), §10.7, eqs. (10.71)–(10.75) and the welfare discussion, pp. 497–498; Hosios (1990); Rogerson, Shimer and Wright (2005).

Checked: `check-10-16-2-9.py` verifies with sympy, for an arbitrary $M(U,V)$, the identity $W'(V) - [q(1-\phi)y - c] = q\, y\, [\eta - (1-\phi)]$ and the externality split; for $M = UV/(1+U+V)$ it checks η, $M_{VV} < 0$ and non-constant returns, then solves (F) and (P) by root-finding with $U=1$, $y=8$, $c=1$: $\phi = \frac{1}{2}$ gives $V^{EQ} = V^* = 2$; $\phi = \frac{1}{4}$ gives $V^{EQ} = 4 > V^*$;

$\phi = 0.6$ gives $V^{EQ} = 1.2 < V^*$; at $U = 3$ the efficient $\phi \approx 0.59$, not $\frac{1}{2}$. The solution manual also answers yes; it does not state where η is evaluated or the direction of the inefficiency.

Romer 2.9 — A closed-form solution of the Ramsey model

Statement in the book, p. 95. Background: Romer §2.2–2.6.

Step — Model and maintained assumptions. Romer's Ramsey model in units of effective labor: no depreciation, $f(k) = k^\alpha$ with $0 < \alpha < 1$, household utility weighted by $L(t)/H$, and $\beta \equiv \rho - n - (1-\theta)g > 0$ (eq. (2.2)); take $n + g > 0$. The equations of motion are

$$\frac{\dot c}{c} = \frac{f'(k) - \rho - \theta g}{\theta} \quad (2.24), \qquad \dot k = f(k) - c - (n+g)k \quad (2.25).$$

With $\theta = \alpha$ and $f'(k) = \alpha k^{\alpha-1}$, (2.24) becomes

$$\frac{\dot c}{c} = \frac{\alpha k^{\alpha-1} - \rho - \alpha g}{\alpha} = k^{\alpha-1} - \frac{\rho + \alpha g}{\alpha}. \qquad (\star)$$

(a)

Step — $\dot c = 0$. With $c > 0$, $(\star)$ vanishes iff $\alpha k^{*\,\alpha-1} = \rho + \alpha g$, i.e. $k^{*\,1-\alpha} = \alpha/(\rho + \alpha g)$. Raising both sides to the power $1/(1-\alpha)$:

$$k^* = \Big(\frac{\alpha}{\rho + \alpha g}\Big)^{1/(1-\alpha)}.$$

(b)

Step — $\dot k = 0$. (2.25) at k^* gives $c^* = k^{*\alpha} - (n+g)k^* = k^*\big[k^{*\,\alpha-1} - (n+g)\big]$. Substitute $k^{*\,\alpha-1} = (\rho + \alpha g)/\alpha$ from (a):

$$c^* = k^*\Big[\frac{\rho + \alpha g}{\alpha} - n - g\Big] = \frac{\rho - \alpha n}{\alpha}\, k^* = \frac{\rho - \alpha n}{\alpha}\Big(\frac{\alpha}{\rho + \alpha g}\Big)^{1/(1-\alpha)}.$$

It is positive: expanding, $\beta + (1-\alpha)(n+g) = \rho - n - (1-\alpha)g + (1-\alpha)n + (1-\alpha)g = \rho - \alpha n$, so $\rho - \alpha n > 0$. Define $x^* \equiv c^*/k^* = (\rho - \alpha n)/\alpha$.

(c)

Step — $\dot z$. Since $y = k^\alpha$, $z = k/y = k^{1-\alpha}$ and $k^{\alpha-1} = 1/z$. By the chain rule and (2.25),

$$\begin{aligned}\dot z &= (1-\alpha)k^{-\alpha}\dot k = (1-\alpha)k^{-\alpha}\big[k^\alpha - c - (n+g)k\big] = (1-\alpha)\Big[1 - \frac{c}{k}\,k^{1-\alpha} - (n+g)\,k^{1-\alpha}\Big] \\ &= (1-\alpha)\big[1 - (x + n + g)\,z\big].\end{aligned}$$

Step — $\dot x/x$. $\ln x = \ln c - \ln k$, so $\dot x/x = \dot c/c - \dot k/k$. From $(\star)$, $\dot c/c = 1/z - (\rho + \alpha g)/\alpha$; from (2.25), $\dot k/k = k^{\alpha-1} - c/k - (n+g) = 1/z - x - n - g$. Hence

$$\frac{\dot x}{x} = \frac{1}{z} - \frac{\rho}{\alpha} - g - \frac{1}{z} + x + n + g = x - \frac{\rho - \alpha n}{\alpha} = x - x^*.$$

The $1/z$ terms cancel only because $\theta = \alpha$: the marginal product raises consumption growth by $f'(k)/\theta = (\alpha/\theta)k^{\alpha-1}$, the average product raises capital growth by $k^{\alpha-1}$.

(d)

Step — The conjecture is the saddle path. If x is constant, $\dot x = x(x - x^*) = 0$ with $x > 0$ forces $x(t) = x^*$ for all t: the saddle path is the ray $c = x^*k$. No other path is admissible. Let $u \equiv 1/x$; then $\dot u = -\dot x/x^2 = -(x - x^*)/x = x^*u - 1$, so $u(t) = 1/x^* + [u(0) - 1/x^*]\,e^{x^*t}$. If $x(0) > x^*$, u reaches 0 at $T = \ln[x(0)/(x(0) - x^*)]/x^*$: $c/k \to \infty$ in finite time, so k is driven to zero while $c > 0$ — the paths above the saddle path that Romer rules out in §2.3 (at $k = 0$, c would have

to jump to zero, violating (2.24)). If $x(0) < x^*$, $u \to \infty$ and $x \to 0$; by (c), $z \to 1/(n+g)$, so $f'(k) = \alpha/z \to \alpha(n+g) < n+g$ and $e^{-R(s)}e^{(n+g)s}k(s) \to \infty$: households would not exhaust lifetime wealth, the paths below the saddle path ruled out in §2.3 using (2.15).

Step — (i) Path of z. With $x = x^*$, use $x^* + n + g = (\rho - \alpha n + \alpha n + \alpha g)/\alpha = (\rho + \alpha g)/\alpha = 1/z^*$, where $z^* \equiv {k^*}^{1-\alpha} = \alpha/(\rho + \alpha g)$. Then

$$\dot z = (1-\alpha)\Big[1 - \frac{z}{z^*}\Big] = -\lambda\,(z - z^*), \qquad \lambda \equiv \frac{1-\alpha}{z^*} = \frac{(1-\alpha)(\rho+\alpha g)}{\alpha}.$$

Since $\frac{d}{dt}\big[e^{\lambda t}(z - z^*)\big] = e^{\lambda t}\big[\dot z + \lambda(z - z^*)\big] = 0$, we have $e^{\lambda t}[z(t) - z^*] = z(0) - z^*$, that is,

$$z(t) = z^* + \big[z(0) - z^*\big]e^{-\lambda t}.$$

Step — (ii) Path of y. $y = k^\alpha = (k^{1-\alpha})^{\alpha/(1-\alpha)} = z^{\alpha/(1-\alpha)}$ and $z(0) = k(0)^{1-\alpha}$, so

$$y(t) = \Big\{{k^*}^{1-\alpha} + \big[k(0)^{1-\alpha} - {k^*}^{1-\alpha}\big]e^{-\lambda t}\Big\}^{\alpha/(1-\alpha)}, \qquad {k^*}^{1-\alpha} = \frac{\alpha}{\rho + \alpha g},$$

with $k(t) = z(t)^{1/(1-\alpha)}$, $c(t) = x^* k(t)$ and $y^* = {z^*}^{\alpha/(1-\alpha)} = {k^*}^\alpha$.

Step — Speed of convergence. Let $a \equiv \alpha/(1-\alpha)$, so $y = z^a$, and use $|y - y^*|$ so the log is defined on either side of y^*. With $\dot y = a z^{a-1}\dot z = -\lambda a z^{a-1}(z - z^*)$,

$$\frac{d\ln|y - y^*|}{dt} = \frac{\dot y}{y - y^*} = -\lambda\,\psi(z), \qquad \psi(z) \equiv \frac{a z^{a-1}(z - z^*)}{z^a - {z^*}^a}.$$

As $z \to z^*$, $(z^a - {z^*}^a)/(z - z^*) \to a{z^*}^{a-1}$, so $\psi \to 1$: the asymptotic speed is λ. But ψ is constant only if $a = 1$, i.e. $\alpha = \theta = \frac{1}{2}$ (then $y = z$ and the speed is exactly λ). For $\alpha < \frac{1}{2}$, z^a is concave: below the balanced growth path ($z < z^*$) the slope $a z^{a-1}$ exceeds the chord slope $(z^{*a} - z^a)/(z^* - z)$, so $\psi > 1$; above it, $\psi < 1$. With $\alpha = \frac{1}{3}$, $a = \frac{1}{2}$ and $\psi = \frac{1}{2}z^{-1/2}(z^{1/2} + z^{*1/2}) = \frac{1}{2}(1 + y^*/y)$: an economy starting at $y = y^*/2$ converges at 1.5λ and slows to λ. So the answer is **no**: the rate is constant for z, not for y.

Step — Link to the linearization of §2.6. With $\theta = \alpha$, $f''(k^*)c^*/\theta = (\alpha - 1){k^*}^{\alpha-1}x^* = -\lambda x^*$ (using ${k^*}^{\alpha-1} = 1/z^*$ and $(1-\alpha)/z^* = \lambda$). Then (2.36) at $\mu = -\lambda$ reads $\lambda^2 + \beta\lambda - \lambda x^* = \lambda(\lambda + \beta - x^*)$, and

$$\alpha(\lambda + \beta - x^*) = (1-\alpha)(\rho + \alpha g) + \alpha\rho - \alpha n - \alpha(1-\alpha)g - \rho + \alpha n = 0.$$

So Romer's negative root μ_1 in (2.39) equals $-\lambda$ exactly, and the linearized saddle path ($\dot{\tilde k} = \beta\tilde k - \tilde c = \mu_1\tilde k$, i.e. $\tilde c = (\beta + \lambda)\tilde k = x^*\tilde k$) is the exact ray $c = x^* k$. The linearization is exact for z; its error for y comes only from the curvature of $y = z^a$.

(e)

Step — Equation (2.25). On the conjectured path $k = z^{1/(1-\alpha)}$, so $\dot k = \frac{1}{1-\alpha}z^{\alpha/(1-\alpha)}\dot z = z^{\alpha/(1-\alpha)}\big(1 - z/z^*\big) = z^{\alpha/(1-\alpha)} - z^{1/(1-\alpha)}/z^*$. The right side of (2.25), using $c = x^* k$ and $x^* + n + g = 1/z^*$, is $k^\alpha - (x^* + n + g)k = z^{\alpha/(1-\alpha)} - z^{1/(1-\alpha)}/z^*$. They coincide.

Step — Equation (2.24). With x^* constant, $\dot c/c = \dot k/k = z^{\alpha/(1-\alpha)-1/(1-\alpha)} - 1/z^* = 1/z - 1/z^* = k^{\alpha-1} - (\rho + \alpha g)/\alpha$, which is $(\star)$, i.e. (2.24) with $\theta = \alpha$.

Step — Conclusion. Both equations of motion hold at every t, the path starts at the given $k(0)$, and $(k, c) \to (k^*, c^*)$, so $f'(k) \to \rho + \theta g$ and $e^{-R(s)}e^{(n+g)s}k(s)$ eventually falls at rate $\beta > 0$: (2.15) holds with equality. **Yes** — the conjecture is the equilibrium path, and by the argument in (d) it is the only one.

Economic intuition. Consumption growth loads on the average product y/k with weight α/θ (because $f'(k) = \alpha y/k$); capital growth loads on it with weight 1. With $\theta = \alpha$ the weights coincide, so c and k grow at the same rate whenever $c/k = x^*$: households consume a fixed fraction x^* of their capital.

The capital-output ratio then obeys a linear equation and converges at the constant rate λ; output inherits a varying rate because $y = z^{\alpha/(1-\alpha)}$ is curved.

Where this goes wrong. Adding depreciation (Romer's chapter 2 has none); writing $\rho + g$ instead of $\rho + \theta g$ in the Euler equation; and reading the constant rate of z as a constant rate for y.

Ref: Romer (2012), eqs. (2.2), (2.15), (2.24)–(2.25), (2.36)–(2.39); §2.3 (saddle path) and §2.6 (speed of adjustment); Smith (2006).

Checked: `check-10-16-2-9.py` re-derives $\dot{z}$ and $\dot{x}/x$ from (2.24)–(2.25) with sympy; checks k^*, c^*, the $u = 1/x$ solution and blow-up time, that the closed-form path satisfies (2.24)–(2.25) identically, and that (2.39) with $\theta = \alpha$ gives $\mu_1 = -\lambda$ and saddle slope x^*; integrates (2.24)–(2.25) numerically from $c(0) = x^* k(0)$ ($\alpha = \theta = \frac{1}{3}$, $\rho = 0.04$, $n = 0.01$, $g = 0.02$, $k(0) = k^*/2$) and matches $z(t)$ to the closed form; and confirms $\psi = \frac{1}{2}(1 + y^*/y) > 1$ for $\alpha = \frac{1}{3}$ and $\psi \equiv 1$ for $\alpha = \frac{1}{2}$. The solution manual agrees on (a)–(e); it calls the speed "clearly not constant" without the $\alpha = \frac{1}{2}$ exception, and leaves $x = x^*$ a conjecture checked only through (e).

Romer 2.12 — Temporary taxes on investment income

Statement in the book, p. 96. Background: Romer §2.6–2.7, problems 2.10–2.11.

Notation of Romer ch. 2: k and c per unit of effective labor, no depreciation, CRRA utility with coefficient θ, and $\rho - n - (1-\theta)g > 0$. Investment income is taxed at $\tau(t) \in \{0, \tau\}$, $0 < \tau < 1$, and the revenue is rebated lump-sum (the setting of 2.10). Problems 2.10–2.11 are solved only as far as 2.12 needs them.

Step — The two systems of dynamics (2.10a). A household holding k_h has the flow budget $\dot{k}_h = (1-\tau(t))f'(k)k_h + w + T - c - (n+g)k_h$, with wage $w = f(k) - kf'(k)$ and a transfer $T = \tau(t)f'(k)k$ that does not depend on its own k_h. Its Euler equation is Romer's (2.20) with the after-tax return $r(t) = (1-\tau(t))f'(k(t))$; imposing $k_h = k$ in the budget,

$$\frac{\dot{c}}{c} = \frac{(1-\tau(t))f'(k) - \rho - \theta g}{\theta},$$
$$\dot{k} = (1-\tau)f'(k)k + \big[f(k) - kf'(k)\big] + \tau f'(k)k - c - (n+g)k = f(k) - c - (n+g)k.$$

The tax drops out of the resource constraint, so the $\dot{k} = 0$ locus $c = \bar{c}(k) \equiv f(k) - (n+g)k$ does not move. The $\dot{c} = 0$ locus moves from $f'(k^*) = \rho + \theta g$ to $f'(k_\tau) = (\rho + \theta g)/(1-\tau) > \rho + \theta g$; as $f'' < 0$, $k_\tau < k^*$: the vertical line shifts left. Call N the untaxed system (steady state $E = (k^*, c^*)$, saddle path S_N) and T the taxed one ($E_\tau = (k_\tau, c_\tau)$, saddle path S_T). Since $\rho + \theta g > n + g$, $f'(k^*) > n + g = f'(k_{GR})$, so $k^* < k_{GR}$, $\bar{c}$ is increasing on $(0, k^*]$ and $c_\tau < c^*$.

Step — Regions and traps. Under T, $\dot{c} < 0$ iff $k > k_\tau$, and $\dot{k} < 0$ iff $c > \bar{c}(k)$. Let

$$A = \{k > k_\tau,\ c > \bar{c}(k)\}, \qquad B = \{k > k_\tau,\ c < \bar{c}(k)\}, \qquad C = \{k < k_\tau,\ c > \bar{c}(k)\}:$$

in A both variables fall; in B, c falls and k rises; in C, c rises and k falls. B and C are traps. In B, k rises, so k_τ is never reached, and on the locus the motion is straight down, back into B. In C, k falls, and c rises while $\bar{c}(k)$ falls with k, so neither boundary is ever reached. A_N, B_N, C_N are the same regions under N, with k^* in place of k_τ. Along S_N, c and k move together toward E, so S_N is increasing; it lies strictly below $\bar{c}$ for $k < k^*$ (there $\dot{k} > 0$) and above c^* for $k > k^*$.

Step — What can jump, and when (2.11b). k is a stock with finite $\dot{k}$, so it is continuous. For c, Romer's (2.18) gives $\theta \ln c(t) = \ln B - \ln \lambda - \beta t + R(t) - (n+g)t$, with $R(t) = \int_0^t r(s)\,ds$. R is continuous even at the dates where r jumps (the start or end of the tax), so for a given λ the path of c is continuous. The multiplier λ, the marginal utility of initial wealth, is set when the plan is made and changes only when news arrives. Here the only news date is 0, so c can jump at 0 and nowhere else. In words: a foreseen upward jump at t' would let the household gain a first-order amount of utility by moving consumption from just after t' to just before it, at a negligible cost in interest.

Step — Where the economy must be when the last regime starts. After the last change of regime, N governs forever and Romer's §2.3 argument applies: above S_N, k hits zero in finite

time and c would then have to jump down (ruled out above); below S_N, $c \to 0$ while k rises past k_{GR}, and households hold capital whose present value never vanishes, violating the transversality condition. So at the date of the last change the economy must be on S_N. Neither k nor c jumps at that foreseen date, so the dynamics of the preceding regime must carry it onto S_N exactly then (S_N is not invariant under T, so arriving early does not help). With a permanent tax the target is S_T instead: in 2.10(b) c jumps at 0 straight onto S_T; in 2.11 it jumps by less at 0, drifts northwest under N and reaches S_T exactly at t_1.

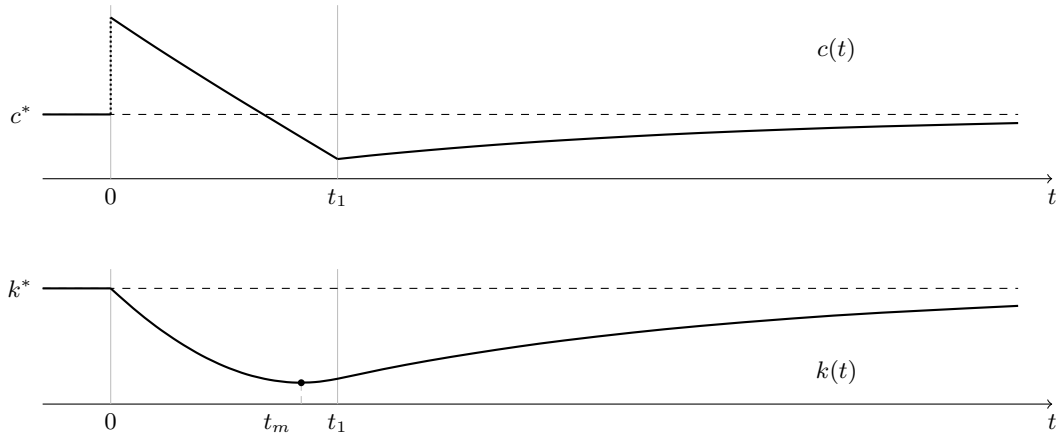
(a)

Step — Regimes. T on $[0, t_1)$, N from t_1 on, news only at 0. So c jumps at 0 to some c_0, both variables are continuous at t_1, and the T-trajectory from (k^*, c_0) must be on S_N exactly at t_1.

Step — The jump is upward. If $c_0 < c^*$, the point (k^*, c_0) is in B; if $c_0 = c^*$, it is on the locus with $\dot{k} = 0 > \dot{c}$ and enters B at once. Either way the path is trapped in B, with $k(t) > k^*$ and $c(t) < c^*$ for all $t > 0$, while S_N lies above c^* wherever $k > k^*$: S_N is never reached. Hence $c_0 > c^*$.

Step — Dynamics while the tax is in force. (k^*, c_0) is in A, so c and k both start to fall. If the path stayed in A until t_1, it would be above $\bar{c}$ at some $k < k^*$, where S_N lies below $\bar{c}$; if it entered the trap C, it would stay above $\bar{c}$ for good. So it must cross the $\dot{k} = 0$ locus into B at some $t_m < t_1$, at $k_m \in (k_\tau, k^*)$; from then on c keeps falling while k rises. At t_1, $c(t_1) < c(t_m) = \bar{c}(k_m) < \bar{c}(k^*) = c^*$, and since S_N is increasing, $k(t_1) < k^*$. A path starting on or above S_T cannot cross S_T and never enters B, so c_0 lies strictly between c^* and the height of S_T at k^*: the jump is smaller than under the permanent tax of 2.10(b). By continuity some c_0 in that interval arrives on S_N exactly at t_1 (the arrival time tends to 0 as $c_0 \downarrow c^*$ and to infinity as c_0 approaches S_T).

Step — After the tax is removed. The economy rides S_N up to E: c and k both rise. At t_1 the $\dot{c} = 0$ line jumps back from k_τ to k^*, so $\dot{c}$ switches from negative to positive (a kinked trough in c), while $\dot{k} = f(k) - c - (n + g)k$ is continuous (k is smooth). In sum: c jumps up at 0, falls throughout the tax period to below c^* and then rises back to c^*; k falls to a minimum at $t_m < t_1$, is already rising when the tax ends, and returns to k^*.



(a): tax on $[0, 10)$. Simulated paths ($f = k^{1/3}$, $\rho = 0.04$, $\theta = 2$, $n = 0.01$, $g = 0.02$, $\tau = 0.3$), plotted as $100 \ln(c/c^*)$ and $100 \ln(k/k^*)$; dotted: the jump at 0.

Economic intuition. While the tax is in force the after-tax return is below $\rho + \theta g$, so the Euler equation makes consumption fall; once it is lifted, capital is scarce ($k < k^*$), the return exceeds $\rho + \theta g$ and consumption grows back. With no long-run change and no jump allowed at t_1, consumption must start the tax period above c^* and end it below: households consume more and invest less while saving is penalized, and start rebuilding capital before t_1 because they know the high untaxed return is coming. The revenue is rebated, so the resource constraint is unchanged and the whole effect works through the after-tax return; it is weaker than for a permanent tax because saving is penalized only for a while.

(b)

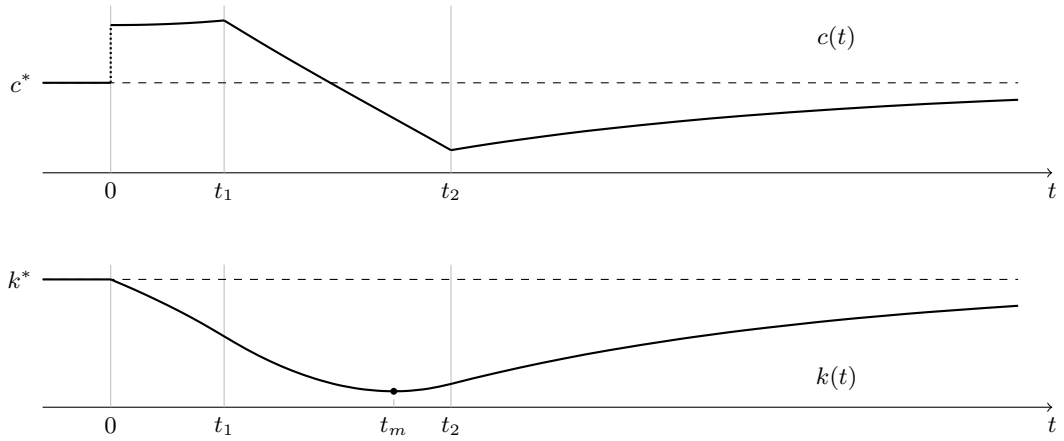
Step — Regimes. N on $[0, t_1)$, T on $[t_1, t_2)$, N from t_2 on, news only at 0. So c jumps only at 0, is continuous at t_1 and t_2, and the economy must reach S_N exactly at t_2. Before t_1 the untaxed Euler equation governs c: the $\dot{c} = 0$ line is still at k^*.

Step — The jump is upward. If $c_0 = c^*$, the economy rests at E until t_1 and then faces the situation ruled out in (a). If $c_0 < c^*$, (k^*, c_0) is in B_N, a trap of N, so at t_1 it has $k > k^*$ and $c < c^*$: it is then in the trap B of T and never reaches S_N, which lies above c^* for $k > k^*$. Hence $c_0 > c^*$.

Step — Between the announcement and t_1. At (k^*, c_0), $\dot{c} = 0$ and $\dot{k} = c^* - c_0 < 0$; once $k < k^*$, $\dot{c} > 0$. The path is in C_N, a trap of N: c rises and k falls until t_1. This is an explosive path of N (on its own it would drive k to zero); it is the equilibrium path only because the regime changes before it explodes.

Step — While the tax is in force. At t_1 the $\dot{c} = 0$ line jumps to k_τ while (k, c) is continuous. If $k(t_1) \leq k_\tau$, the point would be in (or enter) C, a trap above $\bar{c}$, and S_N would never be reached. So $k(t_1) > k_\tau$, the point is in A, and the argument of (a) applies with t_2 in place of t_1: c falls throughout (t_1, t_2); k keeps falling until the path crosses the locus at $t_m \in (t_1, t_2)$ and rises afterwards; at t_2 the economy is on S_N with $c < c^*$ and $k < k^*$.

Step — After the tax ends. The economy rides S_N up to E. In sum: c jumps up at 0, rises until t_1 (a kinked peak, as $\dot{c}$ turns negative), falls during the tax period to below c^* (a kinked trough at t_2) and returns to c^*; k falls from 0 to a minimum at $t_m \in (t_1, t_2)$ and then rises smoothly back to k^*.



(b): tax on $[5, 15)$, announced at 0. Same calibration and units as in (a).

Economic intuition. Anticipated policy moves the economy before it is implemented. Households know that saving will be poorly rewarded during $[t_1, t_2)$, so consumption must fall over that window (Euler). Since c cannot jump at t_1, the high level from which it falls must be built up beforehand, and under the untaxed Euler equation consumption grows only if $f'(k) > \rho + \theta g$, i.e. only if capital is already below k^*. So the economy starts to decumulate at the announcement, and consumption is front-loaded relative to the tax period.

Where this goes wrong. (i) Letting c jump when the tax starts or ends: both dates are known at 0. (ii) In (a), sending the economy to S_T: that answers the permanent tax of 2.10(b); with a temporary tax the target is S_N at t_1, and the economy is never on S_T. (iii) Claiming that k falls during the whole tax period: the landing point on S_N lies below the $\dot{k} = 0$ locus, so k is already rising when the tax ends. (iv) Shifting the $\dot{k} = 0$ locus: it stays put because the revenue is rebated; it would shift down if the revenue financed government purchases (2.10f).

Ref: Romer (2012), §2.2–2.3 (eqs. (2.17)–(2.18), (2.24)–(2.25), the saddle path), §2.6 (an unanticipated change: jump onto the new saddle path), §2.7 (temporary versus permanent changes); problems 2.10–2.11, pp. 95–96.

Checked: `check-2-12-11-1.py` confirms symbolically that the rebate leaves $\dot{k}$ unchanged, then simulates the calibrated economy by shooting c_0 for 2.10(b), 2.11 and both parts here, and asserts every claim above: $c^* < c_0$, with jumps $100 \ln(c_0/c^*)$ of 11.3 (2.10b), 7.5 (a) and 4.5 (b); the monotonicity of c on each interval and the sign switch of $\dot{c}$ at the policy dates; the minimum of k at $t_m \approx 8.4 < t_1 = 10$ in (a) and at $t_m \approx 12.5 \in (5, 15)$ in (b); and arrival on

S_N (traced separately by backward integration) exactly at the last switch, with $c < c^*$, $k < k^*$. Romer's solution manual (pp. 2-18–2-19) reaches the same answer for both parts.

Romer 11.1 — The forward-looking Cagan model

Statement in the book, p. 576. Background: Romer §11.1.

Money demand as given in the problem, with $b > 0$ and c a constant: real balances $m_t - p_t$ fall when expected inflation $E_t p_{t+1} - p_t$ rises, because with output and the real rate fixed, higher expected inflation raises the nominal rate (the Fisher identity (11.3)) and with it the cost of holding money. Write $\phi \equiv b/(1+b) \in (0,1)$, so that $1 - \phi = 1/(1+b)$. Nothing below uses more than the linearity of E_t and the law of iterated projections, so E_t can be read as perfect foresight ($E_t x_{t+i} = x_{t+i}$) throughout.

(a)

Step — Collect the terms in p_t. Expanding the right side, $m_t - p_t = c - bE_t p_{t+1} + bp_t$, so $(1+b)p_t = m_t - c + bE_t p_{t+1}$ and

$$p_t = \frac{1}{1+b}(m_t - c) + \frac{b}{1+b} E_t p_{t+1} = (1-\phi)(m_t - c) + \phi E_t p_{t+1}.$$

Today's price level is a weighted average of $m_t - c$ (the price level that would clear the market with zero expected inflation) and the price level expected for tomorrow.

(b)

Step — Shift one period and take expectations. (a) holds at every date, so $p_{t+1} = (1-\phi)(m_{t+1} - c) + \phi E_{t+1} p_{t+2}$. Applying E_t, which is linear,

$$E_t p_{t+1} = (1-\phi)(E_t m_{t+1} - c) + \phi E_t\big[E_{t+1} p_{t+2}\big] = (1-\phi)(E_t m_{t+1} - c) + \phi E_t p_{t+2},$$

by the law of iterated projections: everything known at t is still known at $t+1$, so $E_t[E_{t+1} x] = E_t x$; today's forecast of tomorrow's forecast is today's forecast (a predictable revision would mean today's forecast is not rational).

(c)

Step — Iterate. Substituting (b) into (a) gives $p_t = (1-\phi)(m_t - c) + \phi(1-\phi)(E_t m_{t+1} - c) + \phi^2 E_t p_{t+2}$. By induction, for every $n \geq 1$,

$$p_t = (1-\phi) \sum_{i=0}^{n-1} \phi^i \left(E_t m_{t+i} - c\right) + \phi^n E_t p_{t+n}, \qquad E_t m_t = m_t.$$

For $n = 1$ this is (a). If it holds for n, apply (a) at $t+n$ and take E_t with the law of iterated projections, $E_t p_{t+n} = (1-\phi)(E_t m_{t+n} - c) + \phi E_t p_{t+n+1}$; substituting turns the last term into $(1-\phi)\phi^n(E_t m_{t+n} - c) + \phi^{n+1} E_t p_{t+n+1}$, which is the formula for $n+1$.

Step — Impose no bubbles. The problem's condition is $\lim_{n\to\infty} \phi^n E_t p_{t+n} = 0$. Letting $n \to \infty$ and using $(1-\phi)\sum_{i\geq 0} \phi^i = (1-\phi)/(1-\phi) = 1$ (valid as $0 < \phi < 1$),

$$p_t = -c + \sum_{i=0}^{\infty} \frac{b^i}{(1+b)^{i+1}} E_t m_{t+i},$$

provided expected money grows by less than the factor $1/\phi = (1+b)/b$ per period. The weights are positive and sum to one: $p_t + c$ is a weighted average of current and expected future money stocks.

Step — Why the condition is needed. (a) alone does not pin down p_t. If p_t^F is the solution above and B_t satisfies $E_t B_{t+1} = B_t/\phi$, then $p_t^F + B_t$ also satisfies (a): the extra terms are B_t on the left and $\phi E_t B_{t+1} = B_t$ on the right. Such a bubble is expected to grow at rate $1/b$ per period, pushing expected inflation up and real balances down without end (Romer's footnote 1, p. 516). Since $\phi^n E_t B_{t+n} = B_t$ for all n, the no-bubbles condition says exactly $B_t = 0$.

(d)

Step — The chain of expectations. From (c), $\partial p_t / \partial E_t m_{t+i} = b^i/(1+b)^{i+1} > 0$. The mechanism runs backward in time. Money-market equilibrium at $t+i$ makes the expected price level $E_t p_{t+i}$ higher. Then expected inflation from $t+i-1$ to $t+i$ is higher, so the nominal rate at $t+i-1$ is higher and real money demand at $t+i-1$ is lower; with m_{t+i-1} unchanged, p_{t+i-1} must rise to shrink real balances. That raises expected inflation from $t+i-2$ to $t+i-1$, and so on back to t: higher expected inflation from t to $t+1$ lowers the demand for real balances today, and since m_t is given, only a higher p_t clears the money market. Each step back passes on the fraction $\phi = b/(1+b)$ of the effect, so it decays geometrically with the horizon, and more slowly the more interest-elastic money demand is (larger b). If the whole expected path of money rises by one unit, p_t rises by one unit, since the weights sum to one.

Economic intuition. The price level behaves like an asset price: it is the present value of current and future fundamentals (the money stocks), discounted at rate $1/b$, since $1/(1+1/b) = \phi$. News about future money moves today's price level before any money is printed, just as news about future dividends moves today's share price.

(e)

Step — Substitute the path. With $E_t m_{t+i} = m_t + gi$ in the formula of (c),

$$p_t = -c + (1-\phi)\sum_{i=0}^{\infty} \phi^i (m_t + gi) = m_t - c + g\,(1-\phi)S, \qquad S \equiv \sum_{i=0}^{\infty} i\,\phi^i.$$

Step — Evaluate the series. The $i=0$ term of S is zero, so $S = \sum_{j\geq 1} j\phi^j$, while $\phi S = \sum_{i\geq 0} i\phi^{i+1} = \sum_{j\geq 1}(j-1)\phi^j$. Subtracting, $S - \phi S = \sum_{j\geq 1}\phi^j = \phi/(1-\phi)$, so $S = \phi/(1-\phi)^2$ and $(1-\phi)S = \phi/(1-\phi) = \dfrac{b/(1+b)}{1/(1+b)} = b$. Hence

$$p_t = m_t - c + bg.$$

Step — Check. The same formula one period ahead gives $E_t p_{t+1} = m_t + g - c + bg$, so expected inflation is $E_t p_{t+1} - p_t = g$ and $m_t - p_t = c - bg = c - b(E_t p_{t+1} - p_t)$: money demand holds. The no-bubbles condition holds too, since $\phi^n(m_t + gn - c + bg) \to 0$.

Step — Effect of g. $\partial p_t/\partial g = b > 0$. With m_t given, faster expected money growth raises expected inflation one for one, raises the nominal rate (the Fisher effect) and lowers the demand for real balances by b per unit of g; the money market clears today only if p_t jumps up by $b\,\Delta g$, and real balances $m_t - p_t = c - bg$ fall. This is the discrete-time counterpart of Romer's Figure 11.2: when a higher money growth rate becomes known, the price level jumps although the money stock does not.

Economic intuition. What sets today's price level is the expected path of money, not just its current stock. Run in reverse, the same result says that a credible disinflation (a fall in g) lowers p_t on impact unless m_t jumps up by $b\,|\Delta g|$, which is why the ends of hyperinflations come with bursts of money growth while prices stabilize (Romer, pp. 517–518).

Where this goes wrong. (i) Shifting (a) forward but keeping E_t inside: the shifted equation holds with $E_{t+1}p_{t+2}$, and only the law of iterated projections turns $E_t[E_{t+1}p_{t+2}]$ into $E_t p_{t+2}$. (ii) Mishandling the constant: each term carries $-(1-\phi)\phi^i c$, and these add up to exactly $-c$. (iii) Reading $\partial p_t/\partial g = b$ as an effect that arrives with the extra money: it is immediate, through expected inflation and money demand, with m_t unchanged.

Ref: Romer (2012), §11.1, pp. 514–518 (eq. (11.3), Figure 11.2, footnote 1 on bubble paths); the no-bubbles condition parallels problem 8.8.

Checked: `check-2-12-11-1.py` re-derives (a) with sympy and verifies: the iterated formula of (c) over eight forward substitutions; the weights $b^i/(1+b)^{i+1}$ and their unit sum; the series $\sum_i i\phi^i = \phi/(1-\phi)^2$; the closed form $p_t = m_t - c + bg$, by symbolic summation and by a 3,000-term numerical sum; that it clears the money market and satisfies the no-bubbles condition; $\partial p_t/\partial g = b$; and that adding a bubble with $E_t B_{t+1} = B_t/\phi$ preserves (a). Romer's solution manual

(pp. 11-1–11-2) agrees: $p_t = m_t - c + bg$ and $\partial p_t/\partial g = b$.

Romer 11.22 — Cagan with adaptive expectations

Statement in the book, p. 583. Background: Romer §11.9.

Step — Environment. Notation of §11.9: $m \equiv M/P$ (real balances), $g_M \equiv \dot{M}/M$, $\pi \equiv \dot{P}/P$; S^{MAX} is the largest seignorage that can be collected in a steady state. In our words, the two assumptions are: real balances always equal Cagan's demand evaluated at *expected* inflation, $m = Ce^{-b\pi^e}$ $(C, b > 0)$; expectations move toward actual inflation at speed β, $\dot{\pi}^e = \beta(\pi - \pi^e)$, with $0 < \beta < 1/b$. As in the text, real purchases G are financed entirely by money creation, so $S(t) = G$ at every date.

(a)

Step — Real balances inherit the motion of expectations. Taking logs, $\ln m(t) = \ln C - b\pi^e(t)$; differentiating with respect to time,

$$\frac{\dot{m}(t)}{m(t)} = -b\,\dot{\pi}^e(t). \tag{1}$$

Step — The financing constraint. Seignorage is $S = \dot{M}/P = (\dot{M}/M)(M/P) = g_M m$, so $S = G$ gives $g_M = G/m$. Since $m = M/P$, $\dot{m}/m = g_M - \pi$; hence, using (1),

$$\pi(t) = g_M(t) - \frac{\dot{m}(t)}{m(t)} = \frac{G}{m(t)} + b\,\dot{\pi}^e(t). \tag{2}$$

At a given instant $\pi^e(t)$ is predetermined, and so is $m(t) = Ce^{-b\pi^e(t)}$; the two unknowns are $\pi(t)$ and $\dot{\pi}^e(t)$, linked by (2) and by the adaptive rule.

Step — Solve the two equations. Substituting (2) into $\dot{\pi}^e = \beta(\pi - \pi^e)$,

$$\dot{\pi}^e = \beta\Big[\frac{G}{m} + b\,\dot{\pi}^e - \pi^e\Big] \quad\Longrightarrow\quad (1 - b\beta)\,\dot{\pi}^e = \beta\Big[\frac{G}{m} - \pi^e\Big].$$

Because $b\beta < 1$ we can divide by $1 - b\beta > 0$; substituting $m = Ce^{-b\pi^e}$ then gives

$$\begin{aligned}\dot{\pi}^e(t) &= \frac{\beta}{1 - b\beta}\,\frac{G - \pi^e(t)\,m(t)}{m(t)} \\ &= \frac{\beta}{1 - b\beta}\Big[\frac{G}{C}\,e^{b\pi^e(t)} - \pi^e(t)\Big] \equiv \Phi\big(\pi^e(t)\big).\end{aligned} \tag{3}$$

Step — Actual inflation. From the adaptive rule, $\pi = \pi^e + \dot{\pi}^e/\beta$; with (3),

$$\pi(t) = \pi^e(t) + \frac{G/m(t) - \pi^e(t)}{1 - b\beta} = \frac{G/m(t) - b\beta\,\pi^e(t)}{1 - b\beta}. \tag{4}$$

Step — The same law of motion as (11.75). Write $\pi^e = (\ln C - \ln m)/b$ and combine (1) with (3):

$$\frac{\dot{m}}{m} = -b\,\dot{\pi}^e = \frac{b\beta}{1 - b\beta}\,\frac{1}{m}\Big[\frac{\ln C - \ln m}{b}\,m - G\Big],$$

which is exactly (11.75). The two models are one model: in §11.9 define $x \equiv (\ln C - \ln m)/b$, the inflation rate that would make current holdings desired; then (11.71) reads $\dot{x} = \beta(\pi - x)$, an adaptive rule for x. Figures 11.9–11.10 are thus the phase diagrams of this problem in $(\ln m, \dot{m}/m)$ space; since $(\ln m, \dot{m}/m) = (\ln C - b\pi^e,\ -b\dot{\pi}^e)$, the diagrams in $(\pi^e, \dot{\pi}^e)$ below are those figures rotated by 180° and rescaled.

Where this goes wrong. The book asks for $\dot{\pi}^e$ "as a function of $\pi(t)$". Read literally, the adaptive rule already is one; the analogue of (11.75) gives the motion of the state variable in terms of the state alone, which is (3) (the solution manual reads it the same way). π cannot serve as the argument: by (4), $d\pi/d\pi^e = b\,(g_M - \beta)/(1 - b\beta)$ changes sign where $g_M = \beta$, so two values of π^e map into

the same π. Dropping the term $b\dot{\pi}^e$ in (2), i.e. setting $g_M = \pi$ off the steady state, loses the factor $1/(1-b\beta)$ and, with it, the role of $\beta < 1/b$.

(b)

Step — The sign of $\dot{\pi}^e$ is the sign of a seignorage gap. In a steady state $\dot{\pi}^e = 0$ gives $\pi = \pi^e$ and $\dot{m} = 0$ gives $g_M = \pi$, so seignorage is $\pi m = C\pi e^{-b\pi} \equiv S(\pi)$, the Laffer curve (11.68). $S'(\pi) = C(1-b\pi)e^{-b\pi}$ vanishes only at $\pi = 1/b$, so $S^{\text{MAX}} = S(1/b) = C/(eb)$. The numerator in (3) is $G - \pi^e m(\pi^e) = G - S(\pi^e)$; since $m > 0$ and $\beta/(1-b\beta) > 0$ (this is where $0 < \beta < 1/b$ enters),

$$\operatorname{sign} \dot{\pi}^e = \operatorname{sign}\big[G - S(\pi^e)\big]. \tag{5}$$

Step — Shape of Φ. Differentiating (3),

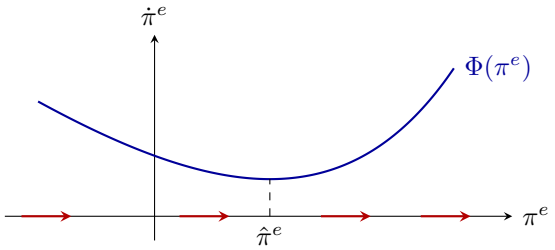
$$\Phi'(\pi^e) = \frac{\beta}{1-b\beta}\Big[\frac{bG}{C}\, e^{b\pi^e} - 1\Big], \qquad \Phi''(\pi^e) = \frac{\beta}{1-b\beta}\,\frac{b^2 G}{C}\, e^{b\pi^e} > 0.$$

Φ is strictly convex, tends to $+\infty$ as $\pi^e \to \pm\infty$, and has its minimum where $\Phi' = 0$, at $\hat{\pi}^e = \ln\big(C/(bG)\big)/b$. There $(G/C)e^{b\hat{\pi}^e} = 1/b$, so

$$\Phi(\hat{\pi}^e) = \frac{\beta}{1-b\beta}\Big[\frac{1}{b} - \frac{1}{b}\ln\frac{C}{bG}\Big] = \frac{\beta}{b(1-b\beta)}\,\ln\frac{ebG}{C} = \frac{\beta}{b(1-b\beta)}\,\ln\frac{G}{S^{\text{MAX}}}. \tag{6}$$

The sign of the minimum is the sign of $G - S^{\text{MAX}}$: this one number separates (b) from (c).

Step — $G > S^{\text{MAX}}$: no steady state. By (6), $\Phi(\pi^e) \geq \Phi(\hat{\pi}^e) > 0$ for every π^e: the curve lies entirely above the axis and every arrow points right. (Drawn for $C = b = 1$, $\beta = \frac{1}{2}$, $G = \frac{1}{2} > S^{\text{MAX}} = e^{-1} \approx 0.37$.)



Step — Dynamics of π^e and m. Since $\dot{\pi}^e \geq \Phi(\hat{\pi}^e) > 0$, $\pi^e(t) \geq \pi^e(0) + \Phi(\hat{\pi}^e)\,t \to \infty$, and $m = Ce^{-b\pi^e}$ falls monotonically toward zero. The fall is not merely asymptotic: multiplying (1) by m and using (3),

$$\dot{m} = -\frac{b\beta}{1-b\beta}\,\big[G - S(\pi^e)\big] \leq -\frac{b\beta}{1-b\beta}\,\big(G - S^{\text{MAX}}\big) < 0,$$

so m reaches zero, and π^e reaches $+\infty$, before $\bar{t} = (1-b\beta)\,m(0)/\big[b\beta\,(G - S^{\text{MAX}})\big]$. Money growth $g_M = G/m$ rises without bound; actual inflation stays above expected inflation, $\pi - \pi^e = \dot{\pi}^e/\beta > 0$, and by (4) explodes with g_M. This is Cagan's hyperinflation.

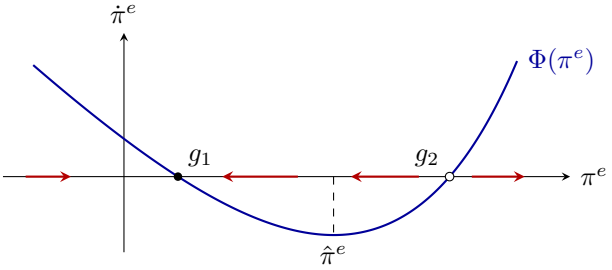
Economic intuition. Because expectations lag, $\pi > \pi^e$ all along, so real balances $Ce^{-b\pi^e}$ exceed the holdings $Ce^{-b\pi}$ that the inflation actually occurring would justify. The government taxes this not-yet-shrunk base, which is how it collects more than S^{MAX}. The gain is transient: expectations catch up, the base shrinks, the required $g_M = G/m$ rises, inflation runs further ahead of expectations, and the process feeds on itself.

(c)

Step — Two steady states. Now $\Phi(\hat{\pi}^e) < 0$ by (6). On $(-\infty, \hat{\pi}^e]$, Φ falls strictly from $+\infty$ to a negative value; on $[\hat{\pi}^e, \infty)$ it rises strictly back to $+\infty$. By continuity it has exactly one zero on each side, $g_1 < \hat{\pi}^e < g_2$. At a zero $S(g_i) = G$, so these are the two money-growth (and inflation)

rates of Figure 11.8, with $g_1 < 1/b < g_2$ (as S rises up to $1/b$ and falls after it), and steady-state real balances are $m_i = Ce^{-bg_i} = G/g_i$.

Step — Stability: a sign argument. $S > G$ strictly between the two roots and $S < G$ outside them (trivially for $\pi^e \le 0$, where $S \le 0 < G$). By (5): $\dot\pi^e > 0$ for $\pi^e < g_1$, $\dot\pi^e < 0$ for $g_1 < \pi^e < g_2$, and $\dot\pi^e > 0$ for $\pi^e > g_2$. Every path that starts below g_2 converges monotonically to g_1: the low-inflation steady state is stable, with basin $(-\infty, g_2)$. Paths that start above g_2 run away to $+\infty$ as in (b): g_2 is unstable. Equivalently, $\Phi'(g_1) < 0 < \Phi'(g_2)$, because Φ is convex and $g_1 < \hat\pi^e < g_2$. Both conclusions rest on $\beta/(1-b\beta) > 0$. (Drawn for $C = b = 1$, $\beta = \frac{1}{2}$, $G = \frac{1}{4}$: $g_1 \approx 0.357$, $g_2 \approx 2.153$.)



Step — Dynamics of m. $m = Ce^{-b\pi^e}$ moves opposite to π^e. From $\pi^e(0) < g_2$, m converges monotonically to $m_1 = G/g_1$, where $g_M = \pi = \pi^e = g_1$. From $\pi^e(0) > g_2$, $S(\pi^e)$ keeps falling as π^e rises, so $\dot m \le -\frac{b\beta}{1-b\beta}[G - S(\pi^e(0))] < 0$ and m collapses in finite time, as in (b).

Economic intuition. Between g_1 and g_2, steady-state revenue at the expected rate exceeds G, so the government needs $g_M = G/m < \pi^e$; by (4) actual inflation falls short of expectations, which are revised down toward g_1. Above g_2 the base erodes faster than the rate rises: the required g_M exceeds π^e, inflation outruns expectations, and the economy slides into hyperinflation. The high-inflation steady state is a threshold, not an outcome, which is why the text focuses on g_1.

Where this goes wrong. If $\beta > 1/b$, the factor $\beta/(1-b\beta)$ in (3) is negative and every arrow reverses: in (b) π^e falls forever (explosive deflation), and in (c) g_2 is stable and g_1 unstable, the "peculiar" case of Romer's footnote 34.

Ref: Romer (2012), §11.9, eqs. (11.67)–(11.75), Figs. 11.8–11.10 and footnotes 33–34; Cagan (1956).

Checked: `check-11-22-6-16.py` re-derives (3) and (4) with sympy from the two equations, confirms that (3) reproduces (11.75) and that $\min\Phi = \frac{\beta}{b(1-b\beta)}\ln(G/S^{\text{MAX}})$; for $C = b = 1$, $\beta = \frac{1}{2}$ it checks $G = \frac{1}{2}$: $\Phi > 0$ on a grid and the simulated π^e passes 30 ($m < 10^{-13}$) at $t \approx 4.75$, before $\bar t \approx 7.57$; and $G = \frac{1}{4}$: $S(g_i) = G$, $\Phi'(g_1) < 0 < \Phi'(g_2)$, paths from below g_1 and from (g_1, g_2) converge to g_1 and the path from above g_2 explodes. The solution manual agrees with (3), with (b) and with the stability ranking in (c), but its text for (c) has a sign typo: for π^e outside $[g_1, g_2]$ it writes $\dot\pi^e < 0$; it must be $\dot\pi^e > 0$, as its own eq. (11) and the arrows in its figure require.

Romer 6.16 — Flexible- and rigid-price firms

Statement in the book, p. 311. Background: Romer §6.5–6.7; Benigno (2015) §4.

Step — Environment. All variables are in logs. Aggregate demand is $y = m - p$ (eq. (6.41) in logs). A firm's desired price is $p^* = p + \phi y$ (eq. (6.58) with the constant normalized to zero), that is, $p^* = (1-\phi)p + \phi m$ (eq. (6.61)), with $\phi > 0$ so that the flexible-price equilibrium is stable (§6.5). Flexible-price firms set $p^f = p^*$ after observing m; rigid-price firms set $p^r = Ep^*$ before. E is the expectation held when rigid prices are set, so a change in m is *anticipated* if it is known at that moment. A fraction $q \in (0,1)$ of firms is rigid: $p = q\,p^r + (1-q)\,p^f$. With every price flexible, $p^* = p$ forces $y = 0$: natural output is $y_n = 0$ and $p = m$.

(a)

Step — Substitute the price index. Inserting $p = q\,p^r + (1-q)\,p^f$ into $p^f = (1-\phi)p + \phi m$,

$$p^f = (1-\phi)\left[q\,p^r + (1-q)\,p^f\right] + \phi m.$$

Step — Collect p^f. Moving the p^f terms to the left,

$$p^f\left[1-(1-\phi)(1-q)\right] = (1-\phi)\,q\,p^r + \phi m, \qquad 1-(1-\phi)(1-q) = q+\phi(1-q) \equiv D > 0.$$

Dividing by D and using $D-\phi = (1-\phi)q$,

$$p^f = \frac{(1-\phi)\,q\,p^r + \phi m}{q+\phi(1-q)} = p^r + \frac{\phi}{q+\phi(1-q)}\,(m-p^r). \tag{A}$$

Flexible-price firms close the fraction ϕ/D of the gap between money and the rigid price. Since $D-\phi = (1-\phi)q$, $\phi/D < 1$ exactly when $\phi < 1$, i.e. when desired prices rise less than one-for-one with the price level ($\partial p^*/\partial p = 1-\phi$: strategic complementarity).

(b)

Step — Expected flexible price. The flexible firms' rule holds whatever m turns out to be, and expectations are linear, so $Ep^f = (1-\phi)\,Ep + \phi\,Em$. The right side is exactly the rigid firms' rule: $Ep^f = p^r$. Each rigid firm expects to charge what the flexible firms will charge.

Step — Expected price level. p^r is chosen with the information behind E, so $Ep^r = p^r$ and

$$Ep = q\,p^r + (1-q)\,Ep^f = q\,p^r + (1-q)\,p^r = p^r.$$

Step — Solve. Substituting $Ep = p^r$ into the rigid firms' rule and dividing by $\phi > 0$,

$$p^r = (1-\phi)\,p^r + \phi\,Em \quad\Longrightarrow\quad \phi\,p^r = \phi\,Em \quad\Longrightarrow\quad p^r = Em. \tag{B}$$

No parameter appears: rigid firms preset the price level that flexible prices would deliver given what they know, and $Ey = Em - Ep = 0$. Check against (A): with $p^r = Em$, $Ep^f = Em + \frac{\phi}{D}(Em-Em) = Em = p^r$, as the first step requires.

(c)

Step — Price level and output. With (B), (A) reads $p^f = Em + \frac{\phi}{D}(m-Em)$, so

$$\begin{aligned} p &= q\,Em + (1-q)\Big[Em + \frac{\phi}{D}(m-Em)\Big] = Em + \frac{(1-q)\,\phi}{D}\,(m-Em), \\ y &= m-p = \Big[1-\frac{(1-q)\,\phi}{D}\Big](m-Em) = \frac{q}{q+\phi(1-q)}\,(m-Em), \end{aligned} \tag{C}$$

because $D-(1-q)\phi = q$.

Step — (i) Anticipated changes. Write $m = Em + u$, where $u \equiv m - Em$ is the part of money unknown when rigid prices were set; by (C), $y = (q/D)\,u$ depends on u alone. An anticipated change raises Em and m by the same Δ and leaves u unchanged: $\Delta y = 0$, while p^r, p^f and p all rise by Δ. Money is neutral because the rigid firms knew the change when they set their prices, so every price incorporates it, and the flexible firms keep their relative price $p^f - p^r = (\phi/D)\,u$ unchanged.

Step — (ii) Unanticipated changes. Raising m by Δ with Em fixed raises u by Δ:

$$\Delta y = \frac{q}{q+\phi(1-q)}\,\Delta > 0, \qquad \Delta p = \frac{(1-q)\,\phi}{q+\phi(1-q)}\,\Delta < \Delta.$$

Since $D-q = \phi(1-q) > 0$, $0 < q/D < 1$. Output absorbs part of the surprise because the fraction q of prices is stuck at Em; by (A), flexible firms raise their prices by $(\phi/D)\Delta$, which is less than Δ when $\phi < 1$.

Step — Mapping to Benigno's AS curve. The AS curve follows from the pricing side alone. From the price index and $Ep = p^r$, $p - Ep = (1-q)(p^f - Ep)$; and $p^f - Ep = (p^f - p) + (p - Ep) = \phi y + (p - Ep)$. Hence

$$q\,(p-Ep) = (1-q)\,\phi\,y \quad\Longrightarrow\quad p - Ep = \frac{(1-q)\,\phi}{q}\,(y-y_n), \qquad y_n = 0. \tag{D}$$

This is Benigno's eq. (17), $p - p^e = \kappa\,(y - y_n)$, under the dictionary: the rigid share q is his share α of prices preset at p^e; $p^r = Ep$ is his p^e; p^f is his $\tilde{p}$, whose rule $\tilde{p} - p = (\sigma^{-1} + \eta)(y - y_n)$ (his eq. (16)) is $p^* - p = \phi y$ with $\phi = \sigma^{-1} + \eta$. Therefore $\kappa = (1-q)\phi/q = (1-\alpha)(\sigma^{-1} + \eta)/\alpha$. Romer's §6.5 economy is the special case $U = C - L^\gamma/\gamma$: utility is linear in C ($\sigma^{-1} = 0$) and $L^\gamma/\gamma = L^{1+\eta}/(1+\eta)$ with $\eta = \gamma - 1$, so $\phi = \gamma - 1 = \sigma^{-1} + \eta$.

Step — Output effect and real rigidity. Combining (D) with the AD curve and $Ep = Em$, $y = (m - Em) - (p - Ep) = (m - Em) - \kappa y$, so

$$y = \frac{m - Em}{1 + \kappa}, \qquad \frac{1}{1+\kappa} = \frac{q}{q + (1-q)\,\phi},$$
$$\frac{\partial}{\partial \phi}\Big(\frac{q}{q + \phi(1-q)}\Big) = -\frac{q\,(1-q)}{\big[q + \phi(1-q)\big]^2} < 0,$$

which reproduces (C). Lower ϕ (more real rigidity) means a flatter AS (lower κ) and a larger output response to the same surprise. Limits: as $\phi \to 0$, $y \to m - Em$, so the whole surprise goes to output although a fraction $1 - q$ of firms is free to reprice (with $p^* = p$ they choose not to); at $\phi = 1$, $p^f = m$ and $y = q\,(m - Em)$; as $\phi \to \infty$, $y \to 0$. With Benigno's calibration ($\alpha = 0.66$, $\sigma = 0.5$, $\eta = 0.2$; his §8), $\kappa = 0.34 \times 2.2/0.66 \approx 1.13$ and a 1% money surprise raises y by $1/(1+\kappa) \approx 0.47\%$; keeping $\alpha = 0.66$ but setting $\phi = 0.2$ gives $\kappa \approx 0.10$ and $\approx 0.91\%$.

Economic intuition. Real rigidity alone is neutral: with $q = 0$, $y = 0$ for every ϕ ($\kappa \to \infty$). It magnifies nominal rigidity: when ϕ is small, flexible firms care mostly about their price relative to the sticky ones, so they track the preset prices rather than money, and a small share of preset prices freezes much of the price level (§6.7). Anticipated money is neutral for any q and ϕ, because no price is stuck at a stale value.

Where this goes wrong. (1) "Anticipated" refers to the information held when rigid prices are set: a change announced after that date is a surprise here even if it is public before m is realized. (2) p^r is a choice, not an outcome to forecast: $Ep^r = p^r$, which is what delivers $Ep = p^r$. (3) Romer's η (the elasticity of demand, Benigno's θ) is not Benigno's η (the inverse Frisch elasticity); and Benigno's $\sigma \equiv \tilde{\sigma} s_c$ is the intertemporal elasticity scaled by the consumption share.

Ref: Romer (2012), §6.5, eqs. (6.41), (6.58), (6.61), and §6.7 (real rigidity); Benigno (2015), §4.1, eqs. (16)–(17), calibration in §8.

Checked: `check-11-22-6-16.py` solves the pricing system with sympy by brute force (without the shortcut $Ep^f = p^r$) and confirms (A), $p^r = Em$, (C), $\partial y/\partial Em = 0$ at given u, $0 < q/D < 1$, (D) with $\kappa = (1-q)\phi/q$, its identity with $(1-\alpha)(\sigma^{-1} + \eta)/\alpha$ under the mapping, $y = (m - Em)/(1+\kappa)$, the sign of $\partial y/\partial \phi$ and the calibration numbers. The solution manual agrees: same p^f, $p^r = Em$, $y = q\,(m - Em)/[\phi + (1-\phi)q]$ and the same real-rigidity derivative (its eq. (19)); it does not discuss Benigno's AS.

Romer 6.10 — Menu costs and multiple equilibria

Statement in the book, pp. 308–309. Background: Romer §6.6–6.7.

Variables are logs. All firms face the same p^*, so a firm that pays the fixed cost Z sets $p_i = p^*$ and one that does not keeps $p_i = 0$. Firms are atomistic: one firm's choice moves neither f nor p, y or p^*. Throughout, $\phi > 0$ (the condition in Romer's footnote 19 for the flexible-price equilibrium to be stable) and $m' \neq 0$ (with $m' = 0$ no firm has a reason to move).

(a)

Step — Eliminate y from the optimal price. Put aggregate demand $y = m' - p$ into $p^* = p + \phi y$:

$$p^* = p + \phi(m' - p) = \phi m' + (1 - \phi)\,p.$$

A firm's desired price is a weighted average of money and the price level, with weight $1 - \phi$ on what the other firms charge.

Step — Impose $p = fp^*$ and solve for p^*. Substituting $p = fp^*$,

$$p^* = \phi m' + (1-\phi) f p^* \implies p^*\big[1 - (1-\phi)f\big] = \phi m' \implies p^* = \frac{\phi m'}{D(f)}, \qquad D(f) \equiv 1 - (1-\phi)f.$$

D is linear in f with $D(0) = 1$ and $D(1) = \phi > 0$, so $D > 0$ on $[0, 1]$ and the division is legitimate.

Step — Price level and output. From $p = fp^*$ and $y = m' - p$,

$$p = \frac{f\phi m'}{1-(1-\phi)f}, \qquad y = m' - \frac{f\phi m'}{D(f)} = m'\,\frac{1-(1-\phi)f-\phi f}{D(f)} = m'\,\frac{1-f}{1-(1-\phi)f}.$$

Checks: $f = 0$ gives $p = 0$, $y = m'$, $p^* = \phi m'$ (output absorbs the whole shock); $f = 1$ gives $p = p^* = m'$ and $y = 0$ (neutrality). In between, $\partial y/\partial f = m'\,[-D(f) + (1-f)(1-\phi)]/D(f)^2 = -\phi m'/D(f)^2$: each extra adjuster pushes output toward its flexible-price level.

(b)

Step — The incentive function. A firm that keeps $p_i = 0$ loses $K(0-p^*)^2$ relative to a firm at p^*; a firm that adjusts loses nothing before paying Z. The gross gain from adjusting, as a function of the fraction f of *other* firms that adjust, is therefore

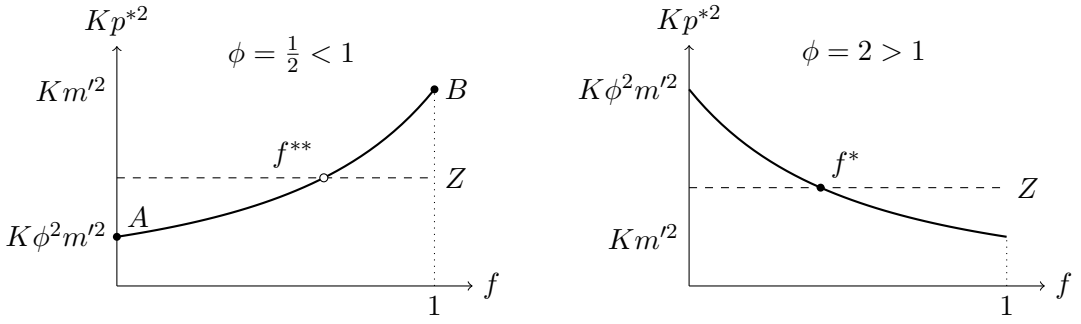
$$G(f) \equiv Kp^{*2} = \frac{K\phi^2 m'^2}{\big[1-(1-\phi)f\big]^2}, \qquad G(0) = K\phi^2 m'^2, \qquad G(1) = \frac{K\phi^2 m'^2}{\phi^2} = Km'^2.$$

Step — Slope and curvature. Write $G = K\phi^2 m'^2 D^{-2}$ with $D'(f) = -(1-\phi)$. By the chain rule,

$$G'(f) = -2K\phi^2 m'^2 D^{-3} D'(f) = \frac{2K\phi^2 m'^2(1-\phi)}{D(f)^3},$$

$$G''(f) = -6K\phi^2 m'^2(1-\phi)D^{-4}D'(f) = \frac{6K\phi^2 m'^2(1-\phi)^2}{D(f)^4} \;\geq 0.$$

As $D > 0$, $\operatorname{sign} G' = \operatorname{sign}(1-\phi)$. *Real rigidity, $\phi < 1$:* G is strictly increasing and convex, rising from $K\phi^2 m'^2$ to Km'^2; price adjustments are strategic complements. *Real flexibility, $\phi > 1$:* G is strictly decreasing and convex, falling from $K\phi^2 m'^2$ to Km'^2; adjustments are strategic substitutes. At $\phi = 1$, $p^* = m'$ whatever the others do and $G \equiv Km'^2$.



$G(f) = Kp^{*2}$ with $K = m' = 1$. Left, $Z = 0.55$: equilibria at A ($f = 0$), B ($f = 1$) and the unstable $f^{**} \approx 0.652$. Right, $Z = 2$: the unique equilibrium $f^* = \sqrt{2} - 1 \approx 0.414$.

(c)

Step — What an equilibrium is. Each firm takes f as given; it adjusts if $G(f) > Z$, stays if $G(f) < Z$, and may do either if $G(f) = Z$. A configuration is an equilibrium when no firm wants to switch:

$$\begin{aligned} f = 1 \text{ is an equilibrium} &\iff G(1) \geq Z \iff Z \leq Km'^2, \\ f = 0 \text{ is an equilibrium} &\iff G(0) \leq Z \iff Z \geq K\phi^2 m'^2, \\ f \in (0,1) \text{ is an equilibrium} &\iff G(f) = Z, \end{aligned}$$

since in the interior case every firm is indifferent, so any split between adjusters and non-adjusters is a best response.

Step — Both corners at once. Both $f = 0$ and $f = 1$ are equilibria iff $K\phi^2 m'^2 \leq Z \leq Km'^2$. This interval is non-empty iff $\phi^2 \leq 1$, i.e. (as $\phi > 0$) iff $\phi \leq 1$, and has positive length iff $\phi < 1$.

Answer: *yes, exactly when* $\phi < 1$ *and* $Z \in [K\phi^2 m'^2,\ Km'^2]$. For Z strictly inside, G crosses Z once more, at $f^{**} = (1-\phi|m'|\sqrt{K/Z})/(1-\phi)$ (solve $G(f) = Z$ as in the last step); this third equilibrium is unstable, since just above it $G > Z$ and more firms want to adjust, and just below it $G < Z$ and adjusters want to stop. At $\phi = 1$ the interval shrinks to $Z = Km'^2$, where $G \equiv Z$ and every $f \in [0,1]$ is an equilibrium.

Step — Neither corner. Neither is an equilibrium iff $G(1) < Z$ and $G(0) > Z$, i.e. $Km'^2 < Z < K\phi^2 m'^2$, which is non-empty iff $\phi > 1$. Answer: *yes, exactly when* $\phi > 1$ *and* $Z \in (Km'^2,\ K\phi^2 m'^2)$.

Step — The interior equilibrium when neither corner works. For $\phi > 1$, G is continuous and strictly decreasing with $G(0) > Z > G(1)$, so exactly one $f^* \in (0,1)$ has $G(f^*) = Z$. Since $D > 0$,

$$\frac{K\phi^2 m'^2}{D(f^*)^2} = Z \iff D(f^*) = \phi|m'|\sqrt{K/Z} \iff 1 + (\phi-1)f^* = \phi|m'|\sqrt{K/Z}$$
$$\iff f^* = \frac{\phi|m'|\sqrt{K/Z} - 1}{\phi - 1},$$

and $0 < f^* < 1 \iff 1 < \phi|m'|\sqrt{K/Z} < \phi \iff Km'^2 < Z < K\phi^2 m'^2$, the same interval. At f^*, $|p^*| = \sqrt{Z/K}$: adjusters move by exactly the amount that makes the menu cost worth paying. With $D^* \equiv D(f^*)$ and $1 - f^* = (\phi - D^*)/(\phi - 1)$, output is $y^* = m'(\phi - D^*)/[(\phi-1)D^*]$, strictly between 0 and m' because $1 < D^* < \phi$: money is partly neutral. The equilibrium is stable (for $f > f^*$, $G < Z$ and adjusters prefer to stay; for $f < f^*$, $G > Z$ and more firms adjust), and with $\phi > 1$ the equilibrium is always unique, because G is strictly monotone and the two corner conditions are incompatible.

	$Z < \min\{G(0), G(1)\}$	Z between $G(0)$ and $G(1)$	$Z > \max\{G(0), G(1)\}$
$\phi < 1$	only $f = 1$	$f = 0$ and $f = 1$ (and f^{**} if Z is interior)	only $f = 0$
$\phi > 1$	only $f = 1$	only f^* (stable)	only $f = 0$

Economic intuition. With $\phi < 1$ the desired price $p^* = \phi m' + (1-\phi)p$ leans on the others' prices. If nobody else moves, a firm's optimal change is only $\phi m'$, worth $K\phi^2 m'^2$; if everybody moves it is the full m', worth Km'^2. A menu cost between the two makes both outcomes self-fulfilling: the same nominal shock leaves the economy either rigid, with $y = m'$ (a recession when $m' < 0$), or flexible, with $y = 0$. Each firm's price change helps the others through aggregate demand, an externality it ignores, which is why Ball and D. Romer (1991) read nominal rigidity as a coordination failure (Romer §6.8). Real rigidity therefore does double duty: it shrinks the private gain from adjusting when others stay put, and it makes adjustment decisions complements. With $\phi > 1$ the others' adjustment shrinks a firm's own desired change, decisions are substitutes, the equilibrium is unique, and an intermediate menu cost yields partial adjustment.

Where this goes wrong. (1) Letting the deviating firm's own choice move p and p^*: with a continuum of firms its deviation has measure zero, so p^* is evaluated at the others' f. (2) Writing the corner conditions with strict inequalities only: an indifferent firm may do either, so the intervals are closed and $f = 1$ survives at $Z = Km'^2$. (3) Reading $G(0) < G(1)$ as multiplicity for every Z: it needs $Z \in [K\phi^2 m'^2, Km'^2]$, and outside that interval the equilibrium is unique even with $\phi < 1$. (4) Treating only expansions: everything depends on m'^2 and $|m'|$, so a contraction is the mirror image.

Ref: Romer (2012), §6.6–6.7 (pp. 275–286): the incentive to adjust, real rigidity and eq. (6.58); §6.8 (from p. 286): coordination failure. Ball and D. Romer (1991), "Sticky Prices as Coordination Failure," *American Economic Review* 81, 539–552.

Checked: `check-6-10-6-7.py` re-derives p^*, p, y, G, G', G'' and f^* with sympy, and with $K = m' = 1$ confirms the interval $[0.25, 1]$ and three equilibria at $Z = 0.55$ for $\phi = \frac{1}{2}$ ($f^{**} = 0.6516$), and the unique $f^* = y^* = \sqrt{2} - 1$ at $Z = 2$ for $\phi = 2$, by scanning $G(f) - Z$ on a grid; the same scan confirms both interval claims, and uniqueness whenever $\phi > 1$, on a grid of Z for $\phi \in \{0.5, 0.75, 1.5, 2\}$. The solution manual agrees on every part; it states the intervals with strict inequalities and does not solve for the interior equilibrium.

Romer 6.7 — The liquidity trap

Statement in the book, p. 308. Background: Romer §6.4; Benigno (2015) §9.

With $\bar{y} \equiv 0$, y is the output gap, i the nominal rate and $i - \pi$ the real rate in the traditional IS curve. The rule (6.26) is a continuously differentiable $r(y, \pi)$ with $r_1 > 0$ and $r_2 > 0$. Two regularity conditions, both true for the linear rule used below, are assumed: for every π the equation $\theta y + r(y, \pi) = 0$ has a solution, and the kink $(\tilde{y}, \tilde{\pi})$ exists (the statement presupposes it). The figures use $r = c + ay + b\pi$ with $\theta = 2$, $a = 1$, $b = 3$, $\lambda = \frac{1}{2}$, and $c = -9$ (then $\tilde{y} > 0$) or $c = 4.5$ (then $\tilde{y} < 0$); every result is proved for a general r.

(a)

Step — Combine the IS curve and the constrained rule. The IS curve says $\theta y = \pi - i$. Substituting $i = \max[0, \pi + r(y, \pi)]$ and using $-\max\{u, v\} = \min\{-u, -v\}$,

$$\theta y = \pi - \max\big[0,\ \pi + r(y, \pi)\big] = \min\big[\pi,\ -r(y, \pi)\big].$$

This single equation is the AD curve. Equivalently, the real rate is $i - \pi = \max[r(y, \pi), -\pi]$: what the bank wants, unless that lies below the floor $-\pi$ imposed by $i \geq 0$.

Step — The two branches. A minimum equals θy iff one term equals θy and the other is not smaller, so a point is on the AD curve iff

$$\begin{aligned} &\text{(L), zero bound binding:} && \theta y = \pi && \text{and} && \pi + r(y, \pi) \leq 0, \\ &\text{(U), zero bound slack:} && \theta y + r(y, \pi) = 0 && \text{and} && \pi + r(y, \pi) \geq 0. \end{aligned}$$

At the kink both hold: $\theta\tilde{y} = \tilde{\pi}$ and $\theta\tilde{y} + r(\tilde{y}, \tilde{\pi}) = 0$.

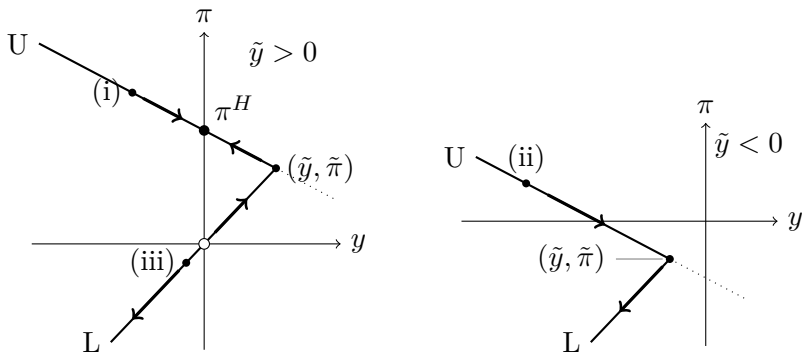
Step — Which branch applies where. On the line $\pi = \theta y$ let $h(y) \equiv \theta y + r(y, \theta y)$; then $h'(y) = \theta + r_1 + \theta r_2 > 0$ and $h(\tilde{y}) = 0$, so a point of the line is on (L) iff $h(y) \leq 0$, i.e. iff $y \leq \tilde{y}$, i.e. iff $\pi \leq \tilde{\pi}$. On (U), $y = Y(\pi)$, the unique root of $\theta y + r(y, \pi) = 0$ (its left side rises in y at rate $\theta + r_1 > 0$), with $Y(\tilde{\pi}) = \tilde{y}$ and, by implicit differentiation, $Y'(\pi) = -r_2/(\theta + r_1) < 0$. Since $r(Y, \pi) = -\theta Y$, the side condition of (U) is $k(\pi) \equiv \pi - \theta Y(\pi) \geq 0$; as $k'(\pi) = 1 + \theta r_2/(\theta + r_1) > 0$ and $k(\tilde{\pi}) = 0$, it holds iff $\pi \geq \tilde{\pi}$. Hence each π has exactly one output level on the AD curve:

$$y = A(\pi) \equiv \begin{cases} \pi/\theta, & \pi \leq \tilde{\pi} \quad \text{(nominal rate at zero)}, \\ Y(\pi), & \pi \geq \tilde{\pi} \quad \text{(rule (6.26) operative)}. \end{cases}$$

Step — Slopes and the kink. In the (y, π) plane, differentiating $\theta y + r(y, \pi) = 0$ and $\theta y = \pi$,

$$\left.\frac{d\pi}{dy}\right|_{\text{U}} = -\frac{\theta + r_1}{r_2} < 0, \qquad \left.\frac{d\pi}{dy}\right|_{\text{L}} = \theta > 0.$$

Above $\tilde{\pi}$ the AD curve slopes down as in Romer's Figure 6.9: lower inflation makes the bank cut the real rate. Below $\tilde{\pi}$ it bends back: with $i = 0$ the real rate is $-\pi$, so lower inflation *raises* it and output falls. A rises on $(-\infty, \tilde{\pi}]$ and falls on $[\tilde{\pi}, \infty)$, so $\tilde{y} = \max_\pi A(\pi)$ is the most output any inflation rate can support. As h is increasing with $h(\tilde{y}) = 0$, $\tilde{y} > 0 \iff h(0) < 0 \iff r(0, 0) < 0$. If π^H is the inflation rate at which the rule is neutral, $r(0, \pi^H) = 0$, then $r_2 > 0$ makes this equivalent to $\pi^H > 0$, a positive nominal rate $i = \pi^H$ at the targeted steady state. With a natural rate in the IS curve, $\theta y = r^n + \pi - i$, and a rule neutral at $r(0, \pi^H) = r^n$, the kink solves $\theta y - r^n + r(y, \theta y - r^n) = 0$, so $\tilde{y} > 0 \iff r(0, -r^n) < r(0, \pi^H) \iff r^n + \pi^H > 0$: $\tilde{y} < 0$ is the slump of Krugman (1998) and Benigno (2015, §9), in which r^n has fallen so far that the targeted steady state would need $i < 0$.



AD curves for the linear rule (dotted: where the rule alone would put AD without the zero bound). Motion along AD follows $\dot{\pi} = \lambda y$: down where $y < 0$, up where $y > 0$. Filled dot: the stable steady state π^H; open dot: the unstable trap $\pi = y = i = 0$. Starting points of cases (i)–(iii) marked.

(b)

Step — A phase line on the AD curve. Inflation is predetermined ($\dot{\pi}$ is finite, so π cannot jump), while y moves instantly to keep the economy on the AD curve. So $y(t) = A(\pi(t))$ and the model reduces to one autonomous equation,

$$\dot{\pi} = F(\pi) \equiv \lambda A(\pi).$$

F is continuous and piecewise C^1, hence locally Lipschitz: the solution is unique, π moves monotonically, and it never crosses a zero of F. The zeros are the steady states, $A(\pi) = 0$: on (L), $\pi/\theta = 0$ gives $\pi = 0$, admissible iff $0 \leq \tilde{\pi}$, i.e. $\tilde{y} \geq 0$; on (U), $Y(\pi) = 0$ means $r(0, \pi) = 0$, i.e. $\pi = \pi^H$, admissible iff $\pi^H \geq \tilde{\pi}$, which requires $\tilde{y} = Y(\tilde{\pi}) \geq 0$ because Y is decreasing.

Step — A trap no interest-rate rule can escape. Since $\theta y = \min[\pi, -r] \leq \pi$, every $\pi < 0$ gives $y \leq \pi/\theta < 0$ and $\dot{\pi} < 0$, *whatever the rule*: with $i \geq 0$ the real rate is at least $-\pi > 0$.

Step — Steady states and stability. If $\tilde{y} > 0$ and Y turns negative at high inflation (true for the linear rule, and implied by the data of case (i)), there are two steady states, $0 < \tilde{\pi} < \pi^H$, the last inequality because $Y(\tilde{\pi}) = \tilde{y} > 0 = Y(\pi^H)$ with Y decreasing. $F < 0$ on $(-\infty, 0)$; $F > 0$ on $(0, \pi^H)$, because $A = \pi/\theta > 0$ on $(0, \tilde{\pi}]$ and $A = Y > 0$ on $[\tilde{\pi}, \pi^H)$; and $F < 0$ on (π^H, ∞). So π^H is stable with basin $(0, \infty)$, while the liquidity-trap steady state $\pi = y = i = 0$ is unstable; locally, $F'(\pi^H) = -\lambda r_2/(\theta + r_1) < 0 < F'(0) = \lambda/\theta$. If $\tilde{y} < 0$, $A(\pi) \leq \tilde{y} < 0$ for every π and there is no steady state at all. (At $\tilde{y} = 0$ the two merge at the kink, which attracts from above and repels from below.)

Step — Case (i): $\tilde{y} > 0$, $\pi(0) > \tilde{\pi}$, $y(0) < 0$. The economy starts on (U) with $Y(\pi(0)) < 0 < \tilde{y} = Y(\tilde{\pi})$, so $\pi^H \in (\tilde{\pi}, \pi(0))$ by continuity and monotonicity of Y. On (π^H, ∞), $F < 0$: inflation falls monotonically, cannot cross π^H and, being bounded below, converges to a zero of F, which can only be π^H. Output $y = Y(\pi)$ rises monotonically to 0, and the nominal rate stays positive throughout because $\pi > \tilde{\pi}$: an ordinary disinflation, in which the bank cuts the real rate as inflation falls. Near π^H the gap decays at rate $\lambda r_2/(\theta + r_1)$; with the linear rule, $Y(\pi) = -b(\pi - \pi^H)/(\theta + a)$, so exactly $\pi(t) = \pi^H + (\pi(0) - \pi^H)e^{-\mu t}$ and $y(t) = y(0)e^{-\mu t}$, with $\mu = \lambda b/(\theta + a)$.

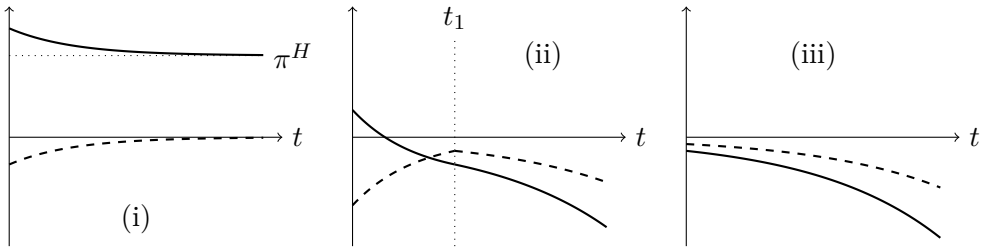
Step — Case (ii): $\tilde{y} < 0$, $\pi(0) > \tilde{\pi}$. Here $y(0) = Y(\pi(0)) \leq \tilde{y} < 0$. *Phase 1, on (U).* $\dot{\pi} = \lambda Y(\pi) \leq \lambda\tilde{y} < 0$, so inflation falls at least at speed $\lambda|\tilde{y}|$ and reaches $\tilde{\pi}$ at some $t_1 \leq (\pi(0) - \tilde{\pi})/(\lambda|\tilde{y}|)$, while output $Y(\pi)$ *rises* toward $\tilde{y}$ because the bank keeps cutting i. *Phase 2, on (L), $t \geq t_1$.* Now $i = 0$, $y = \pi/\theta$ and $\dot{\pi} = (\lambda/\theta)\pi$ with $\pi(t_1) = \tilde{\pi} = \theta\tilde{y} < 0$, so

$$\pi(t) = \tilde{\pi}\, e^{(\lambda/\theta)(t - t_1)} \to -\infty, \qquad y(t) = \tilde{y}\, e^{(\lambda/\theta)(t - t_1)} \to -\infty.$$

Inflation falls for ever; output first recovers, then collapses. With the linear rule, phase 1 is case (i)'s formula aimed at the unreachable $\pi^H = -c/b < \tilde{\pi}$, so $t_1 = \mu^{-1} \ln[(\pi(0) - \pi^H)/(\tilde{\pi} - \pi^H)]$.

Step — Case (iii): $\tilde{y} > 0$, $\pi(0) < \tilde{\pi}$, $y(0) < 0$. The economy starts on (L), so $y(0) = \pi(0)/\theta < 0$ forces $\pi(0) < 0$: it starts below the unstable steady state. For all t, $\dot{\pi} = (\lambda/\theta)\pi$, hence $\pi(t) =$

$\pi(0)e^{(\lambda/\theta)t}$ and $y(t) = \pi(t)/\theta$, both falling without bound with $i = 0$ throughout; π^H exists but is never reached. Had $0 < \pi(0) < \tilde{\pi}$ (so $y(0) > 0$), inflation would have risen, crossed $\tilde{\pi}$ in finite time and converged to π^H along (U): $\pi = 0$ is the watershed between recovery and collapse.



Time paths for the linear rule: $\pi(t)$ solid, $y(t)$ dashed. (i) $c = -9$, $\pi(0) = 4$, so $\pi^H = 3$ and $\mu = \frac{1}{2}$; (ii) $c = 4.5$, $\pi(0) = 1$, $t_1 = 2\ln 5 \approx 3.22$; (iii) $c = -9$, $\pi(0) = -\frac{1}{2}$.

Step — The way out: raise expected inflation. Suppose the bank can make expected inflation exceed current inflation by $\Delta > 0$ through a credible promise of higher future prices. The real rate in the IS curve becomes $i - \pi - \Delta$, the rule still targets it, $i = \max[0, \pi + \Delta + r(y, \pi)]$, and the first step of (a) now gives

$$\theta y = \min\big[\pi + \Delta,\ -r(y, \pi)\big].$$

Branch (U) is unchanged, but (L) shifts out to $y = (\pi + \Delta)/\theta$: the repelling point moves from $\pi = 0$ to $\pi = -\Delta$, and any $\pi > -\Delta$ on (L) now has $y > 0$ and $\dot{\pi} > 0$. In case (iii), a promise $\Delta > -\pi(t)$ turns the spiral into a recovery at $i = 0$ without touching $r(\cdot)$, and it need only last until $\pi > 0$, when the economy has entered the basin $(0, \infty)$ of π^H. In case (ii) no temporary promise is enough, since $A < 0$ at every π: the rule's own steady state, if any, needs $i < 0$. What works there is a permanently higher target, $\pi^H > 0$ (with a natural rate, $\pi^H > -r^n$), which moves the kink to $\tilde{y} > 0$ and creates the basin of recovery, plus a temporary Δ if inflation is already negative.

Economic intuition. Above $\tilde{\pi}$ the rule is stabilising: lower inflation brings a lower real rate, higher output and, through $\dot{\pi} = \lambda y$, an end to disinflation. Below $\tilde{\pi}$ the nominal rate is stuck at zero and the real rate is $-\pi$, so the feedback changes sign: lower inflation raises the real rate, depresses output and lowers inflation further. This deflationary spiral is why $\pi = 0$ is a steady state that repels, and why the rule $r(\cdot)$, however aggressive, is powerless on (L). The lever left is the wedge between expected and current inflation, which is Benigno's (2015, §9) point: in his AD equation (21) the real rate is $i - (\bar{p} - p)$, so raising the long-run price level $\bar{p}$ lowers it even at $i = 0$ and shifts both AD and its zero-bound version AD_0 outward. In Romer's model expected inflation is tied to current inflation, and Δ is exactly the missing $\bar{p}$ channel.

Where this goes wrong. (1) Drawing the zero-bound branch as a continuation of the downward-sloping AD: below $\tilde{\pi}$ the AD bends back. (2) Expecting a more aggressive rule to rescue case (iii): on (L) the rule is inoperative, and for $\pi < 0$ every rule gives $y \le \pi/\theta < 0$. (3) Treating $y(0)$ as a free initial condition: y is pinned by $\pi(0)$ through A, so the conditions on $y(0)$ in (i) and (iii) only place $\pi(0)$ relative to π^H and 0. (4) Claiming that output falls monotonically in (ii): it first rises along (U).

Ref: Romer (2012), §6.4 (pp. 255–267): eqs. (6.22)–(6.23), the rule (6.26) and Figures 6.8–6.9; §11.6 on the zero lower bound. Benigno (2015), §9 ("Liquidity trap") and the AD equation (21), p. 509. Krugman (1998), cited there.

Checked: `check-6-10-6-7.py` confirms the identity $\pi - \max[0, \pi + r] = \min[\pi, -r]$, solves the AD equation by bisection and matches the piecewise $A(\pi)$ (kink $(1, 2)$ and $\pi^H = 3$ for $c = -9$; kink $(-0.5, -1)$ for $c = 4.5$), confirms the branch split at $\tilde{\pi}$ and the slope $-(\theta + r_1)/r_2$ on a nonlinear rule, checks the kink formulas with r^n and with Δ in sympy, and integrates $\dot{\pi} = \lambda A(\pi)$ by RK4: (i) converges to π^H along the closed form, (ii) crosses $\tilde{\pi}$ at $t_1 = 2\ln 5$ with peak output $\tilde{y}$ and then diverges, (iii) diverges, $\pi(0) = 0.1$ recovers, no linear rule tried escapes from $\pi < 0$, and a temporary $\Delta = 1$ rescues case (iii) while $\Delta = 0.3$ does not. The solution manual agrees on (a) and all three cases (it works in (y, r) space); it does not discuss the steady state at $\pi = 0$ or the way out.